MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, 16,384 BIT SCHOTTKY, BIPOLAR, PROGRAMMABLE READ-ONLY MEMORY (PROM), MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, PROM microcircuits which employ thin film nichrome (NiCr) resistors, platinum-silicide, tungsten (W), titanium-tungsten (TiW) or zapped vertical emitter as the fusible link or programming element. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the part number. A special test requirement is included in this specification to screen against devices which may contain excess moisture in the package materials or internal atmosphere (see freeze-out test of 4.2d).

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit	Access times (ns)
01	2048 words/8 bits per word PROM with uncommitted collector	100, 50
02	2048 words/8 bits per word PROM with active pullup and a third high-impedance state output	100, 50
03	2048 words/8 bits per word PROM with uncommitted collector	55, 30
04	2048 words/8 bits per word PROM with active pullup and a third high-impedance state output	55, 30
05	4096 words/4 bits per word PROM with active pullup and a third high-impedance state output	80, 40

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1/2" x 1-1/4"), dual-in-line package
K	F-6 (24-lead, 3/8" x 5/8"), flat package
R	D-8 (20-lead, 1/4" x 1-1/16"), dual-in-line package
L	D-9 (24-lead, 1/4" x 1-1/4"), dual-in-line package
3	C-4 (28-terminal, .450" x .450"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings:

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -10 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}): 1/	
Cases J, L, and R	40°C/W maximum
Case K	60°C/W maximum
Case 3	0.08°C/W maximum 2/
Output voltage range	-0.5 V dc to +V_{CC}
Output sink current	100 mA
Maximum power dissipation (P_D) 3/	1.02 W
Maximum junction temperature (T_J) 4/	+175°C

1.4 Recommended operating conditions:

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output)	8 mA 5/
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. When manufacturer-programmed devices are delivered to the user, an altered item drawing shall be prepared by the contracting activity to specify the required program configuration.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

1/ Heat sinking is recommended to reduce junction temperature.
2/ When a thermal resistance value is included in MIL-M-38510 appendix C, it shall supersede the value stated herein.
3/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
4/ Maximum junction temperature shall not be exceeded except for allowable short circuit duration burn-in screening conditions per method 5004 of MIL-STD-883.
5/ 16 mA for circuits A, B, D, F, H, and I devices.

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3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables.

3.2.2.1 Unprogrammed devices. The truth tables for unprogrammed devices for contracts involving no altered item drawing shall be as specified on figure 2. When required in groups A, B, or C inspection (see 4.4), the devices shall be programmed by the manufacturer prior to test in a checkerboard pattern (a minimum of 50 percent of the total number of bits programmed) or to any altered item drawing pattern which includes at least 25 percent of the total number of bits programmed.

3.2.2.2 Programmed devices. The truth tables for programmed devices shall be as specified by the altered item drawing.

3.2.3 Functional block diagram. The functional block diagram shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics shall be as specified in table I, and shall apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be as specified in table II and, where applicable, by the altered item drawing. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. For programmed devices, the altered item drawing number shall be added to the marking by the programming activity.

3.7 Processing options. Since the PROM is an unprogrammed memory capable of being programmed by either the manufacturer or the user to result in a wide variety of PROM configurations, two processing options are provided for selection in the contract, using an altered item drawing.

3.7.1 Unprogrammed PROM delivered to the user. All testing shall be verified through group A testing as defined in 3.2.2.1 and tables II and III. It is recommended that users perform subgroups 7 and 9 after programming to verify the specific program configuration.

3.7.2 Manufacturer-programmed PROM delivered to the user. All testing requirements and quality assurance provisions herein, including the requirements of the altered item drawing, shall be satisfied by the manufacturer prior to delivery.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 14 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

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a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameter tests prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

d. Freeze-out test. This test shall be conducted as a 100-percent screen on all class S devices having nichrome or platinum silicide as the fusing link. Within no more than 24 hours after completion of burn-in and prior to the final electrical test, all devices having nichrome or platinum silicide as the fusing link shall be subjected to a freeze-out test. If more than 24 hours have elapsed subsequent to the 125°C burn-in exposure, devices shall be conditioned at 125°C for a minimum of 5 hours immediately prior to the freeze-out test. When the freeze-out test is performed, the 25°C final electrical test parameters shall be completed 96 hours after the freeze-out test. The freeze-out test shall be conducted as follows:

(1) Connect devices in the electrical configuration of figure 7 or in the burn-in configuration of figure 4 with the bias cycled, 3 minutes on and 3 minutes off, throughout the duration of test.

(2) Reduce device temperature to T_C = -10°C ±2°C with the bias cycled and maintain at that temperature for a minimum of 5 hours.

(3) With the cycled bias maintained, allow T_C to go to room temperature (by removal from the cold chamber or termination of forced cooling but with no forced heating) and retain for a minimum of 19 hours after the completion of the 5-hour cold soak. T_C shall not exceed 35°C during this period.

(4) Remove bias and subject all devices to subgroup 1 final electrical test to establish continuity of the nichrome or platinum silicide resistors and remove all failed devices from the lot. Count them as screening rejects subject to the PDA requirements of 4.2c.

e. Class B devices processed to an altered item drawing may be programmed either before or after burn-in at the manufacturer's discretion. The required electrical testing shall include, as a minimum, the final electrical tests for programmed devices as specified in table II herein. Class S devices processed by the manufacturer to an altered item drawing shall be programmed prior to burn-in.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Qualification data for subgroups 7 through 11 shall be attributes only.

4.3.1 Qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies to a faster device type which is manufactured identically to a slower device type on this specification, then the slower device type may be part I qualified by conducting only group A electrical tests and any electricals specified as additional group C subgroups and submitting data in accordance with MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -2 mA; V_{IH} = 2.0 V; V_{IL} = 0.8 V	02,04,05	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 8 mA; 2/ V_{IH} = 2.0 V; V_{IL} = 0.8 V	01,02 03,04,05		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -10 mA; T_C = 25°C	01,02 03,04,05		-1.5	V
Maximum collector cut-off current	I_{CEX}	V_{CC} = 5.5 V; V_O = 5.2 V	01,03		100	µA
High-impedance (off-state) output high current	I_{OHZ}	V_{CC} = 5.5 V; V_O = 5.2 V	02,04,05		100	µA
High-impedance (off-state) output low current	I_{OLZ}	V_{CC} = 5.5 V; V_O = 0.5 V	02,04,05		-100	µA
High-level input current	I_{IH}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	01,02 03,04,05		50	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V; V_{IN} = 0.5 V	01,02 03,04,05		-250	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V; V_O = 0.0 V 3/	02,04,05	-10	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V; V_{IN} = 0; outputs = open	01,02 03,04,05		185	mA
Propagation delay time, high-to-low level logic, address to output	t_{PHL1}	V_{CC} = 4.5 V and 5.5 V; C_L = 30 pF (see figure 5)	01,02 03,04 05		100 55 80	ns
Propagation delay time, low-to-high level logic, address to output	t_{PLH1}		01,02 03,04 05		100 55 80	ns
Propagation delay time, high-to-low level logic, enable to output	t_{PHL2}		01,02 03,04 05		50 30 40	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time, low-to-high level logic, enable to output	t_{PLH2}		01,02		50	ns
			03,04		30	
			05		40	

1/ Complete terminal conditions shall be specified in table III.

2/ I_{OL} = 16 mA for circuits A, B, D, F, H, I, and J.

3/ Not more than one output shall be grounded at one time. Output shall be at high logic level prior to test.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) 1/ 2/ 3/	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004) for unprogrammed devices	1*,2,3,7*, 8	1*,2,3,7*, 8
Final electrical test parameters (method 5004) for programmed devices	1*,2,3,7*, 8,9,10,11	1*,2,3,7*, 8,9
Group A test requirements (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8, 9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3,7,8
Group D end-point electrical parameters (method 5005)	1,2,3,7,8	1,2,3,7,8

1/ * PDA applies to subgroups 1 and 7 (see 4.2c).

2/ Any or all subgroups may be combined when using high-speed testers.

3/ Subgroups 7 and 8 shall consist of verifying the pattern specified.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. For unprogrammed devices, a sample shall be selected to satisfy programmability requirements prior to performing subgroups 9, 10, and 11. Twelve devices shall be submitted to programming (see 3.2.2.1). If more than two devices fail to program, the lot shall be rejected. At the manufacturer's option, the sample may be increased to 24 total devices with no more than 4 total device failures allowable.

d. For unprogrammed devices, 10 devices from the programmability sample shall be submitted to the requirements of group A, subgroups 9, 10, and 11. If more than two total devices fail in all three subgroups, the lot shall be rejected. At the manufacturer's option, the sample may be increased to 20 total devices with no more than 4 total device failures allowable.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. For qualification, at least 50 percent of the sample selected for testing in subgroup 5 shall be programmed (see 3.2.2). For quality conformance inspection, the programmability sample (see 4.4.1.c.) shall be included in the subgroup 5 tests.

c. Steady-state life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using a circuit submitted to the qualifying activity for approval. If the alternate burn-in conditions are used, the circuit on figure 4 or equivalent shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration - 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. For qualification, at least 50 percent of the sample selected for testing in subgroup 1 shall be programmed (see 3.2.2). For quality conformance inspection, the programmability sample (see 4.4.1c) shall be included in the subgroup 1 tests.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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4.6 Programming procedure identification. The programming procedure to be utlized shall be identified by the manufacturer's circuit designator. The circuit designator is cross referenced in 6.5 herein with the manufacturer's symbol or FSCM number.

4.7 Programming procedure for circuit A. The programming characteristics of table IVA and the following procedures shall be used for programming the device.

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6A and the programming characteristics of table IVA shall apply to these procedures.

b. Address the PROM with the binary address of the selected word to be programmed. Address inputs are TTL compatible. An open circuit shall not be used to address the PROM.

c. Apply V_{PL} voltage to V_{CC}.

d. Bring the $\overline{CE}_x$ inputs high and the CE_x inputs low to disable the device. The chip enables are TTL compatible. An open circuit shall not be used to disable the device.

e. Disable the programming circuitry by applying a voltage of V_{OPD} to the outputs of the PROM.

f. Raise V_{CC} to V_{PH} with rise time less than or equal to t_{TLH}.

g. After a delay equal to or greater than t_{D1}, apply only one pulse with amplitude of V_{OPE} and duration of t_p to the output selected for programming. Note that the PROM is supplied with fuses intact, which generates an output high. Programming a fuse will cause the output to go low.

h. Lower V_{CC} to V_{PL} following a delay of t_{D2} from programming enable pulse applied to an output.

i. Enable the PROM for verification by applying V_{IL} to $\overline{CE}_x$ and V_{IH} to CE_x.

j. Apply V_{PHV} to V_{CC} and verify bit is programmed.

k. Repeat 4.7a through 4.7j for all other bits to be programmed in the PROM.

l. If any bit does not verify as programmed, it shall be considered a programming reject.

4.8 Programming procedure for circuit B. The programming characteristics of table IVB and the following procedures shall be used for programming the devices:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6B and the programming characteristics of table IVB shall apply to these procedures.

b. Apply V_{IH} to $\overline{CE}_1$ and the binary address of the PROM word to be programmed. Raise V_{CC} to V_{CCP}.

c. After a t_D delay, apply only one V_{OP} to the output to be programmed high. Apply V_{OP} to one output at a time.

d. After a t_D delay, a pulse $\overline{CE}_1$ to a V_{IL} level for a duration of t_p.

e. After t_p and a t_D delay, remove V_{OP} from the programmed output.

f. Other bits in the same word may be programmed sequentially while the V_{CC} input is at the V_{CCP} level by applying V_{OP} pulses to each output to be programmed and pulsing $\overline{CE}_1$ to the V_{IL} level, allowing for proper delays between V_{OP} and $\overline{CE}_1$.

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g. Repeat 4.8b through 4.8e for all bits to be programmed.

h. To verify programming, lower V_{CCP} to V_{CC}. Connect a 10 kilohm resistor between each output and V_{CC}. Apply V_{IL} to $\overline{CE}_1$ input. The programmed outputs should remain in the high state and the unprogrammed outputs should go to the low level.

i. If any bit does not verify as programmed, it shall be considered a programming reject.

4.9 Programming procedures for circuit C, device types 02 and 04. The programming characteristics of table IVC, and the following procedures shall be used for programming device types 02 and 04.

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6C, device types 02 and 04, and the programming characteristics of table IVC, device types 02 and 04, shall apply to these procedures.

b. Terminate all device outputs with a 10 kilohm resistor to V_{CC}. Apply V_{IH} to CE_1.

c. Address the PROM with the binary address of the selected word to be programmed. Raise V_{CC} to V_{CCP}.

d. After a t_D delay (10 µs), apply only one V_{OUT} pulse to the output to be programmed. Program one output at a time.

e. After a t_D delay (10 µs), pulse $\overline{CE}_1$ input to logic "0" for a duration of t_p.

f. After a t_D delay (10 µs), remove the V_{OUT} pulse from the programmed output. (Programming a fuse will cause the output to go to a high-level logic in the verify mode.)

g. Other bits in the same word may be programmed sequentially while the V_{CC} input is at the V_{CCP} level by applying V_{OUT} pulses to each output to be programmed allowing a delay to t_D between pulses as shown on figure 6C.

h. Repeat 4.9b through 4.9g for all other bits to be programmed.

i. To verify programming, after t_D (10 µs) delay, lower V_{CC} to V_{CCH} and apply a logic "0" level to $\overline{CE}_1$ input. The programmed output should remain in the "1" state. Again, lower V_{CC} to V_{CCL} and verify that the programmed output remains in the "1" state.

j. If any bit does not verify as programmed, it shall be considered a programming reject.

4.10 Programming procedure for circuit C, device type 05. The programming characteristics of table IVC, device type 05, and the following procedures shall be used for programming the device.

a. Connect the device in the electrical configuration for programming. The output pins shall be terminated with a 10K ohm resistor to GND and bypass V_{CC} to GND with a 0.01 µF capacitor. The waveforms on figure 6C, device type 05, and the programming characteristics of table IVC, device type 05, shall apply to these procedures.

b. Disable the device by applying V_{IH} to $\overline{CE}_2$ input and V_{IL} to $\overline{CE}_1$. The chip enable pins are TTL compatible.

c. Apply V_{IL} to all other pins.

d. Addressed the PROM with the binary address of the selected word to be programmed and reset T_p = 5 µs. Address inputs are TTL compatible.

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e. After a delay of TD_1, raise the V_{CC} pin to V_{CCp}.

f. After a delay of TD_2, raise the corresponding output pin to V_{OPF}.

g. After a delay of TD_3, lower $\overline{CE}_2$ to V_{IL} for a duration of T_p and simultaneously lower the output to V_{IL} and wait TD_4.

h. Return the $\overline{CE}_2$ to V_{IH}.

i. Wait TD_5 and lower V_{CC} to V_{CCv}.

j. Wait TD_6 and lower $\overline{CE}_2$ to V_{IL} for the duration of T_v.

k. A properly blown fuse will read V_{OL} and unblown fuse will read V_{OH}.

1. If the fuse is blown, go to n.
2. If the fuse is unblown, to to 1.

l. If T_p is less than 30 µs, increment T_p by 5 µs and go to e. If T_p is $\geq$ 5 µs go to m.

m. If T_p is $\geq$ 30 µs, the device is a reject.

n. After a delay of TD_7, select the next output or address to be programmed.

o. Repeat steps 4.10d through 4.10k until all required addresses are programmed.

p. To verify the program keep V_{CC} pin at V_{CCv}. Apply V_{IL} to $\overline{CE}_2$. The programmed fuse will go to the low level and unblown fuse shall remain in the high level.

4.11 Programming procedure for circuit D. The programming characteristics on table IVD and the following procedures shall be used for programming the device.

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6D and the programming characteristics of table IVD shall apply to these procedures.

b. Select the word to be programmed by applying the appropriate voltages to the address pins as well as the required voltages to chip enable pins to select the device.

c. Apply the proper power, V_{CC} = 6.5 V, GND = 0 V.

d. Verify that the bit to be programmed is in the "0" logic state.

e. Enable the chip for programming by application of the chip enable voltage, $V_{p(CE1)}$ = 21.0 V to $\overline{CE}_1$ (pin 20). CE_2 and CE_3 should be left high.

g. Apply I_{OP} programming current ramp to the output to be programmed. The other outputs shall be left open. Only one output may be programmed at a time. During the rise of the current ramp, the required current will be achieved to program the junction. As programming occurs a drop in voltage can be sensed at the output of the device. Upon detection of V_{ps}, the current shall be held for t_{hap} and then shut off.

h. Verify that the programmed bit is in the "1" logic state. Lower $V_{p(CE_1)}$ to 0 V and read the output.

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Note: The PROM is supplied with fuses generating a low-level logic output. Programming a fuse will cause the output to go to a high-level logic in the verify mode.

i. Lower V_{CC} to 0 V. The power supply duty cycle shall be equal to or less than 50 percent.

j. If the bit verifies as not having been programmed at V_{CC} = 6.5 V, then repeat the programming ramp sequence up to 15 times until the bit is programmed. If after 16 programming attempts, the bit does not program, then the device shall be considered a reject.

k. If the bit verifies as having been programmed at V_{CC} = 6.5 V, then one of the following two conditions shall be followed:

(1) If the current required to program was less than $I_{OP}(max)$, then proceed to step l.

(2) If the current required to program was equal to or greater than $I_{OP}(max)$, then the device shall be considered a reject and no further attempts at programming other bits shall be attempted.

l. Repeat 4.11a through 4.11k for all other bits to be programmed.

m. If any bit does not verify as programmed, it shall be considered a programming reject.

4.12 Programming procedure for circuit E. The programming characteristics for this device have been discontinued.

4.13 Programming procedure for circuit F. The programming characteristics on table IVF and the following procedures shall be used for programming the devices:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6F and the programming characteristics of table IVF shall apply to these procedures.

b. Raise V_{CC} to 5.5 V.

c. Address the PROM with binary address of the selected word to be programmed. Address inputs are TTL compatible.

d. Disable the chip by applying V_{IH} to the $\overline{CE}$ inputs and V_{IL} to the CE inputs. The chip enable inputs are TTL compatible.

e. Apply the V_{PP} pulse to the programming pin $\overline{CE}_1$. In order to insure that the output transistor is OFF before increasing voltage on the output pin, the program pins voltage pulse shall precede the output pins programming pulse by T_{D1} and leave after the programming pins programming pulse by T_{D2} (see figure 6F).

f. Apply one V_{OUT} pulse with duration of t_p to the output selected for programming. The outputs shall be programmed one output at a time, since internal decoding circuitry is capable of sinking only one unit of programming current at a time.

Note: The PROM is supplied with fuses generating a high-level logic output. Programming a fuse will cause the output to go to a low-level logic in the verify mode.

g. Other bits in the same word may be programmed sequentially by applying V_{OUT} pulses to each output to be programmed.

h. Repeat 4.13b through 4.13g for all other bits to be programmed.

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i. Enable the chip by applying V_{IL} to the $\overline{CE}$ inputs and V_{IH} to the CE inputs, and verify the program. Verification may check for a low output by requiring the device to sink 12 mA at V_{CC} = 4.2 V and 0.2 mA at V_{CC} = 6.2 V at T_C = 25°C.

j. If any bit does not verify as programmed, it shall be considered a programming reject.

4.14 Programming procedure for circuit G. The programming characteristics of table IVG and the following procedures shall be used for programming:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6G and the programming characteristics of table IVG shall apply to these procedures.

b. Select the desired word by applying high or low levels to the appropriate address inputs. Disable the device by applying a high level to one or more 'active low' chip Enable inputs. NOTE: Address and Enable inputs must be driven with TTL logic levels during programming and verification.

c. Increase V_{CC} from nominal to V_{CCP} (10.5 ±0.5 V) with a slew rate limit of I_{RR} (1.0 to 10.0 V/µs). Since V_{CC} is the source of the current required to program the fuse as well as the I_{CC} for the device at the programming voltage, it must be capable of supplying 750 mA at 11.0 V.

d. Select the output where a logical high is desired by raising that output voltage to V_{OP} (10.5 ±0.5 V). Limit the slew rates to I_{RR} (1.0 to 10.0 V/µs). This voltage change may occur simultaneously with the V_{CC} increase to V_{CCP}, but must not precede it. It is critical that only one output at a time be programmed since the internal circuits can only supply programming current to one bit at a time. Outputs not being programmed must be left open or connected to a high impedance source of 20 kilohms minimum (remember that the outputs of the device are disabled at this time).

e. Enable the device by taking the chip Enable(s) to a low level. This is done with a pulse PWE for 10 µs. The 10 µs duration refers to the time that the circuit (device) is enabled. Normal input levels are used and rise and fall times are not critical.

f. Verify that the bit has been programmed by first removing the programming voltage from the output and then reducing V_{CC} to 5.0 V (±0.25 V). The device must be Enabled to sense the state of the outputs. During verification, the loading of the output must be within specified I_{OL} and I_{OH} limits.

g. If the device is not to be tested for V_{OH} over the entire temperature range subsequent to programming, the verification of step 4.12f is to be performed at a V_{CC} level of 4.0 volts (±0.2 V). V_{OH}, during the 4 volt verification, must be at least 2.0 volts. The 4-volt V_{CC} verification assures minimum V_{OH} levels over the entire temperature range.

h. Repeat 4.14b through 4.14f for each bit to be programmed to a high level. If the procedure is performed on an automatic programmer, the duty cycle of V_{CC} at the programming voltage must be limited to a maximum of 25 percent. This is necessary to minimize device junction temperatures. After all selected bits are programmed, the entire contents of the memory should be verified.

i. If any bit does not verify as programmed, it shall be considered a programming reject.

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4.15 Programming procedure for circuit H. The programming characteristics of table IVH and the following procedures shall be used for programming the devices.

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6H and the programming characteristics of table IVH shall apply to these procedures.

b. Address the word to be programmed, apply 5 volts to V_{CC} and active levels to all chip Enable inputs.

c. Verify the status of a bit location by checking the output level.

d. Decrease V_{CC} to 0 volts.

e. For bit locations that do not require programming, skip steps 4.15f through 4.15l.

f. Increase V_{CC} to $V_{CC(pr)}$ with a minimum current capability of 250 milliamperes.

g. Apply $V_{S(pr)}$ to all chip Enable inputs. $I_I \leq 25$ milliamperes. Active-high enables may be left high.

h. Connect all outputs, except the one to be programmed, to V_{IL}. Only one bit is to be programmed at a time.

i. Apply the output programming pulse for 20 microseconds. Minimum current capability of the programming supply should be 250 milliamperes.

j. After terminating the output pulse, disconnect all outputs from V_{IL} conditions.

k. Reduce the voltage at $\overline{CE}$ input to V_{IL}.

l. Decrease V_{CC} to 0 V.

m. Return to 4.15e until all outputs in the word have been programmed.

n. Repeat 4.15c through 4.15l for each word in memory.

o. Verify programming of every word after all words have been programmed using V_{CC} values of 4.5 and 5.5 volts.

p. If any bit does not verify as programmed, it shall be considered a programming reject.

4.16 Programming procedures for circuit I. The programming characteristics in table IVI and the following procedures shall be used for programming the device:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6I and the programming characteristics of table IVI shall apply to these procedures.

b. Terminate all outputs with a 300-ohm resistor to V_{ONP}. Apply V_{IHP} to the $\overline{CE}2$, CE3, and CE4 inputs and V_{ILP} to the $\overline{CE}1$ inputs.

c. Address the PROM with the binary address of the selected word to be programmed. Raise V_{CC} to V_{CCP}.

d. After a delay of t1, apply only one V_{OP} pulse with a duration of tp, t2 and $d(V_{OP})/dt$ to the output selected for programming. After a delay of t2 and $d(V_{OP})/dt$, pulse $\overline{CE}2$ from V_{IHP} to V_{CEP} for the duration of tp, $2d(V_{CE})/dt$, and t3; $\overline{CE}2$ is then to go to V_{ILP} level.

e. To verify programming after $\overline{CE}1$ has been set to V_{ILP}, lower V_{CC} to V_{CCL}, after a delay of t4. The programmed output should remain in the logic '1' state.

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f. The outputs should be programmed one output at a time, since the internal decoding circuitry is capable of sinking only one unit of programming current at a time. Note that the PROM is supplied with fuses generating a low-level logic output. Programming a fuse will cause the output to go to a high level logic in the verify mode.

g. Repeat 4.16b through 4.16f for all other bits to be programmed.

h. If any bit does not verify as programmed, it shall be considered a programming reject.

4.17 Programming procedures for circuit J. The programming characteristics in table IVJ and the following procedures shall be used for programming the device:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6J and the programming characteristics of table IVJ shall apply to these procedures.

b. Address the PROM with the binary address of the selected word to be programmed. Address inputs are TTL-compatible. An open circuit should not be used to address the PROM.

c. Disable the chip by applying input high (V_{IH}) to the $\overline{CS}$ input. $\overline{CS}$ input must remain at V_{IH} for programming. The chip select is TTL-compatible. An open circuit should not be used to disable the chip.

d. Disable the programming circuitry by applying an Output Voltage Disable of less than V_{OPD} to the output of the PROM. The output may be left open to achieve the disable.

e. Raise V_{CC} to V_{PH} with rise time equal to t_r.

f. After a delay equal to or greater than t_d, apply a pulse with amplitude of V_{OPE} and duration of t_p to the output selected for programming. Note that the PROM is supplied with fuses intact generating an output high. Programming a fuse will cause the output to go low in the verify mode.

g. Other bits in the same word may be programmed while the V_{CC} input is raised to V_{PH} by applying output enable pulses to each output which is to be programmed. The output enable pulses must be separated by a minimum interval of t_d.

h. Lower V_{CC} to 4.5 volts following a delay of t_d from the last programming enable pulse applied to an output.

i. Enable the PROM for verification by applying a logic "0" (V_{IL}) to the $\overline{CS}$ input.

j. Repeat 4.17a through 4.17i for all other bits to be programmed in the PROM.

k. If any bit does not verify as programmed, it shall be considered a programming reject.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

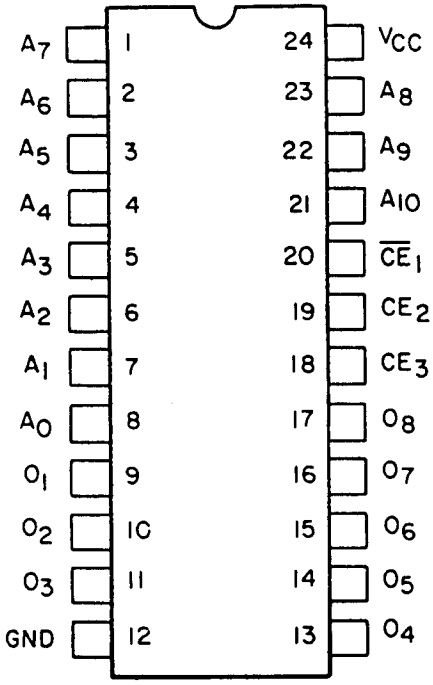
6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

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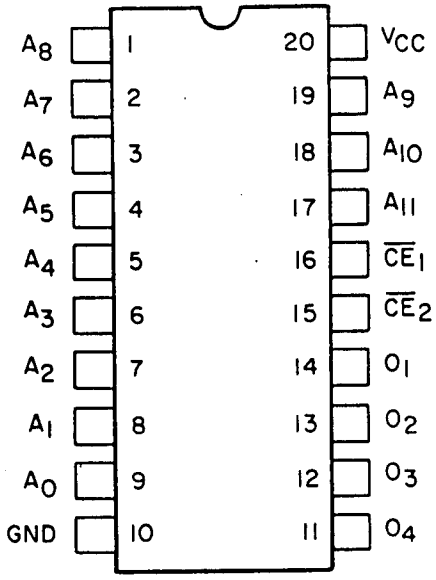
Device types 01, 02, 03, and 04

Cases J, K and L



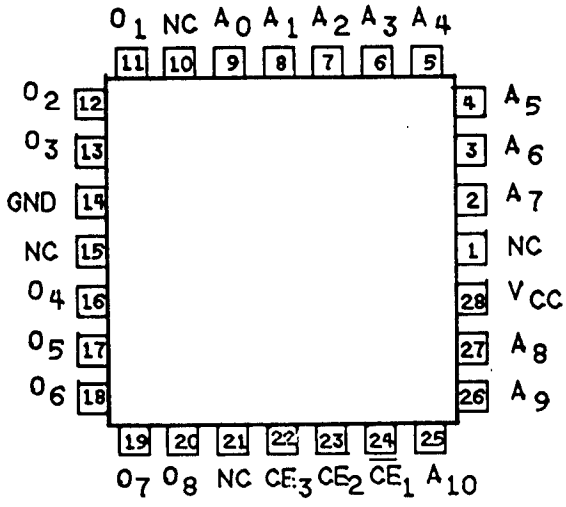
Device type 05

Case R



Device types 02 and 04

Case 3



TOP VIEW

FIGURE 1. Terminal connections.

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Devices types 01, 02, 03, and 04

Word No.	6/ $\overline{CE}_1$	6/ CE_2	6/ CE_3	Address											Data							
				A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_8	O_7	O_6	O_5	O_4	O_3	O_2	O_1
NA	L H	H X	H X	X X	X X	X X	X X	X X	X X	X X	X X	X X	X X	X X	5/ OC	5/ OC	5/ OC	5/ OC	5/ OC	5/ OC	5/ OC	5/ OC

Device type 05

Word No.	6/ $\overline{CE}_1$	6/ $\overline{CE}_2$	Address												Data			
			A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_4	O_3	O_2	O_1
NA	L H	L X	X X	X X	X X	X X	X X	X X	X X	X X	X X	X X	X X	X X	5/ OC	5/ OC	5/ OC	5/ OC

NOTES:
1. NA = Not applicable.
2. X = Input may be high level, low level or open circuit.
3. OC = Open circuit (high resistance output).
4. Program readout can only be accomplished with enable input at low level.
5. The outputs for an unprogrammed device shall be high for circuits A, C (device type 05), E, F, and J and low for circuits B, C (device types 02, 04), D, G, and I.
6. Enable inputs are ANDED.

FIGURE 2. Truth table (unprogrammed).

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Device types 01 and 02
Circuit A

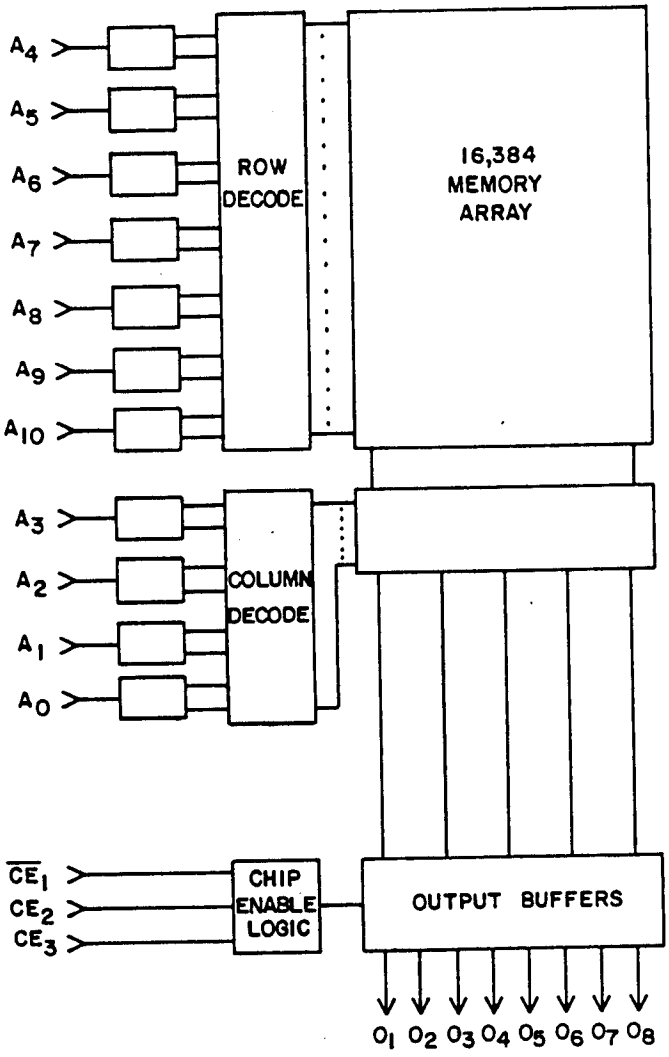


FIGURE 3. Functional block diagrams.

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Device type 05
Circuit A

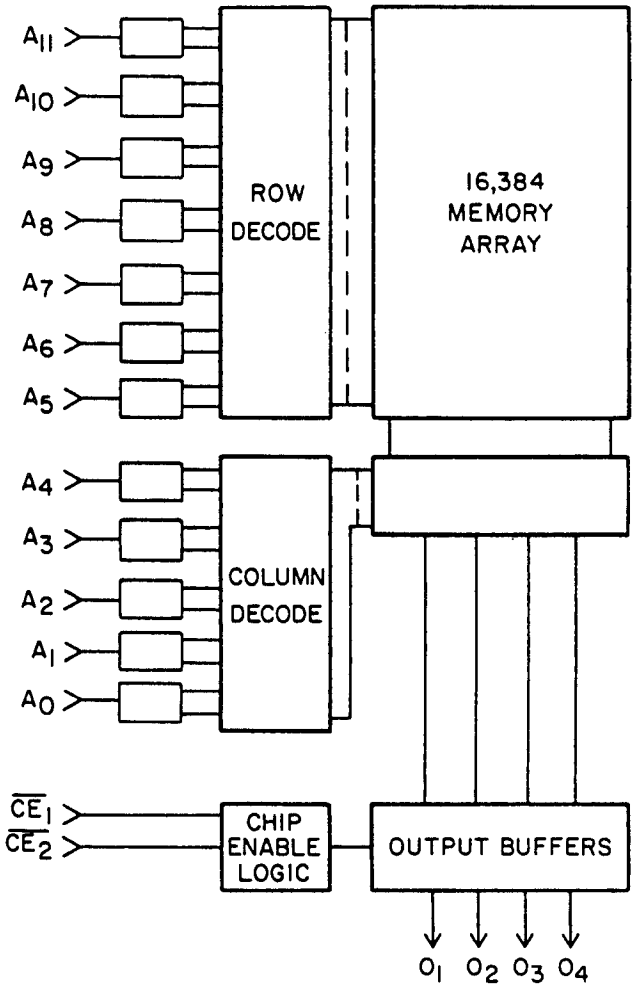


FIGURE 3. Functional block diagrams - Continued.

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Device types 01 and 02
Circuit B

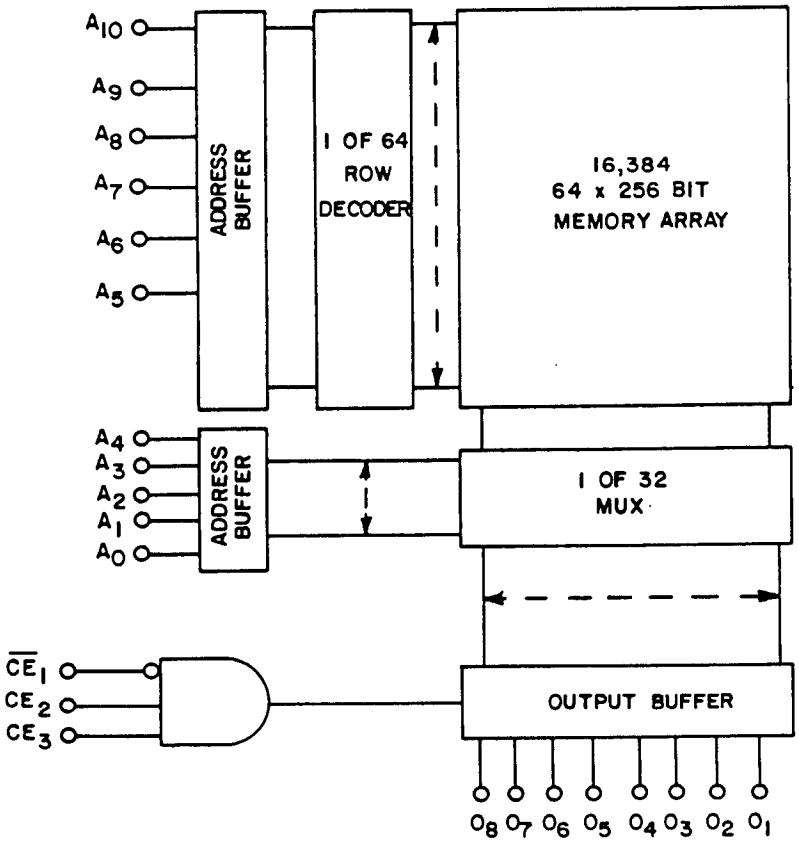


FIGURE 3. Functional block diagrams - Continued.

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Device types 01, 02 and 04
Circuit C

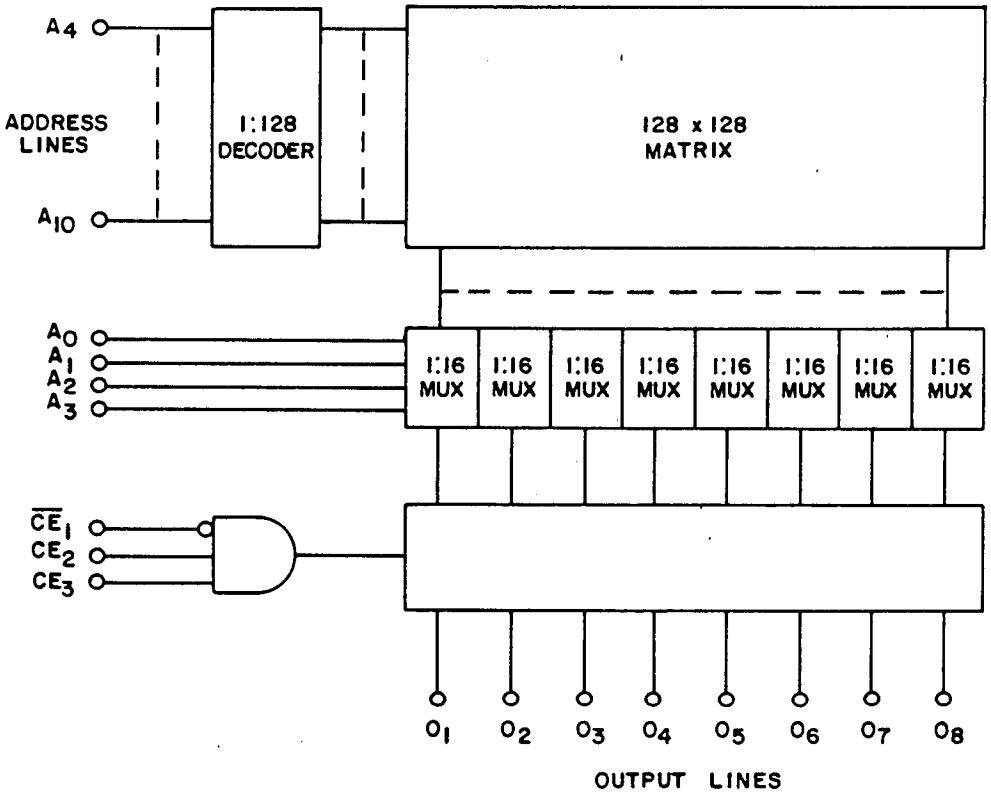


FIGURE 3. Functional block diagrams - Continued.

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Device type 05

Circuit C

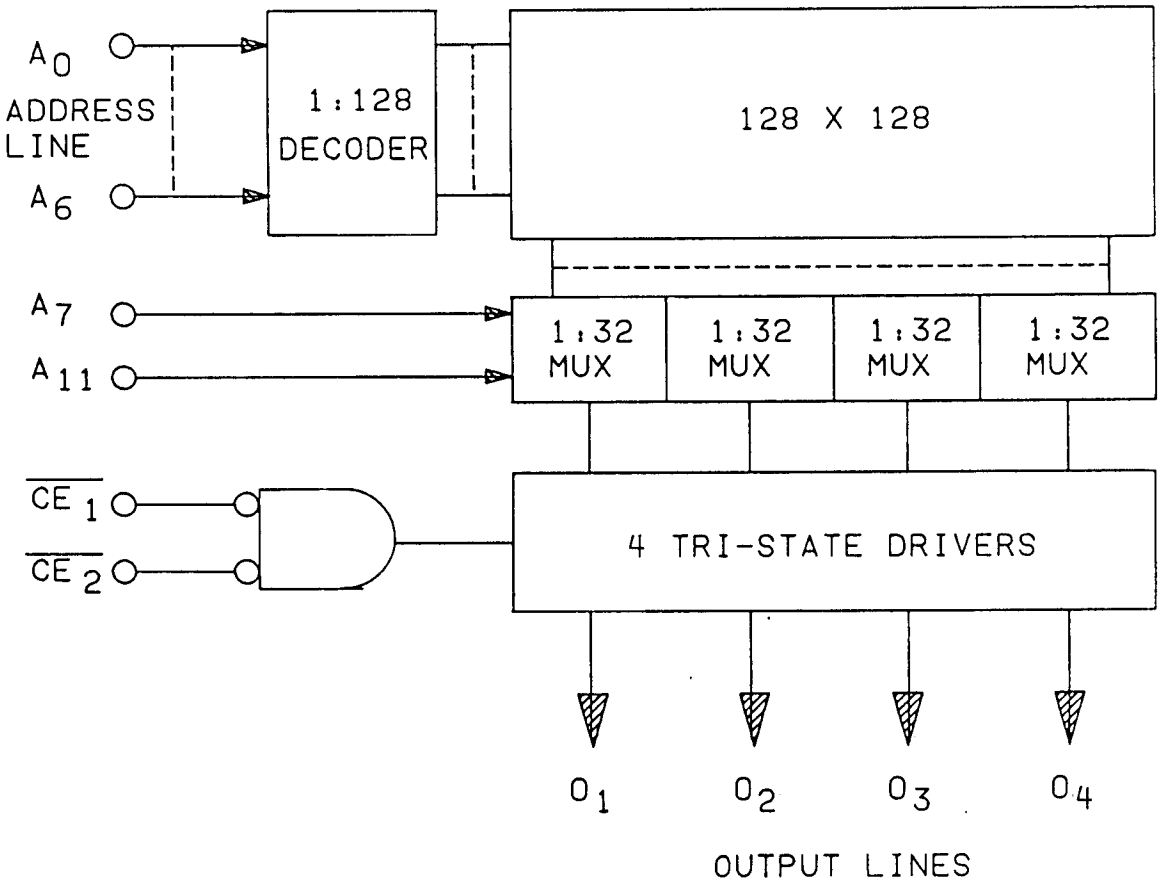


FIGURE 3. Functional block diagrams - Continued.

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Device types 02, 03, and 04
Circuit D

LOGIC DIAGRAM

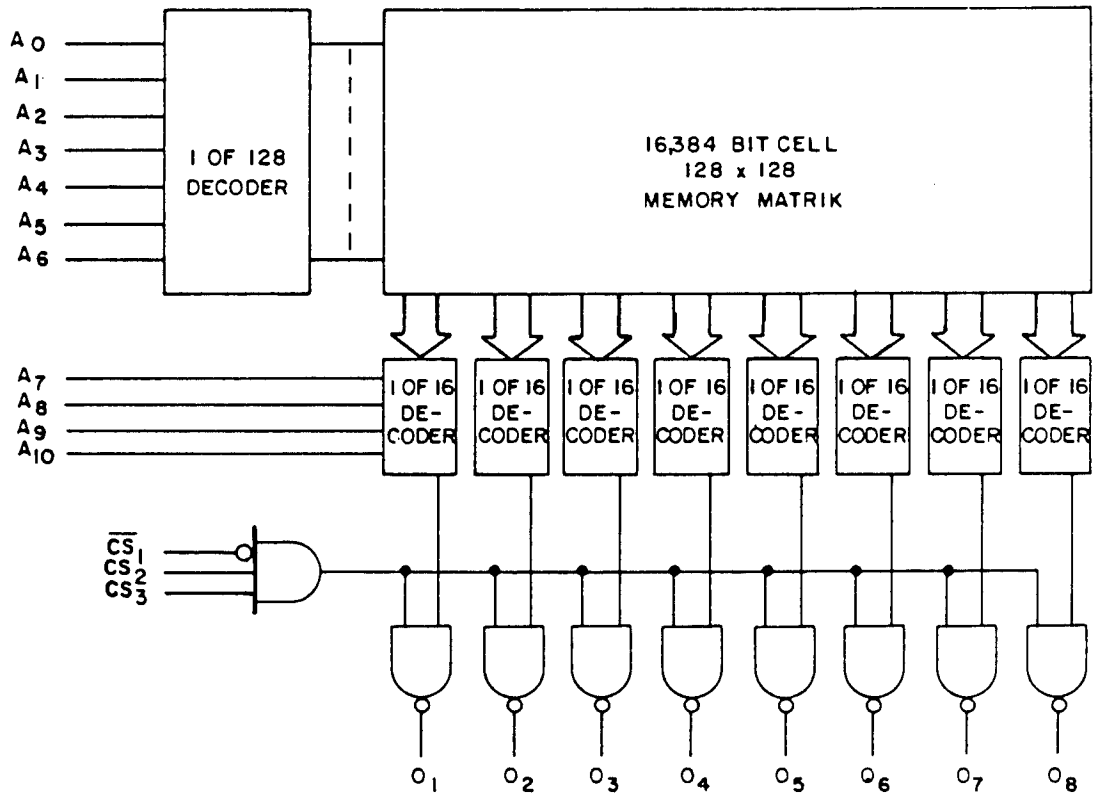


FIGURE 3. Functional block diagrams - Continued.

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Device type 02
Circuits F and I

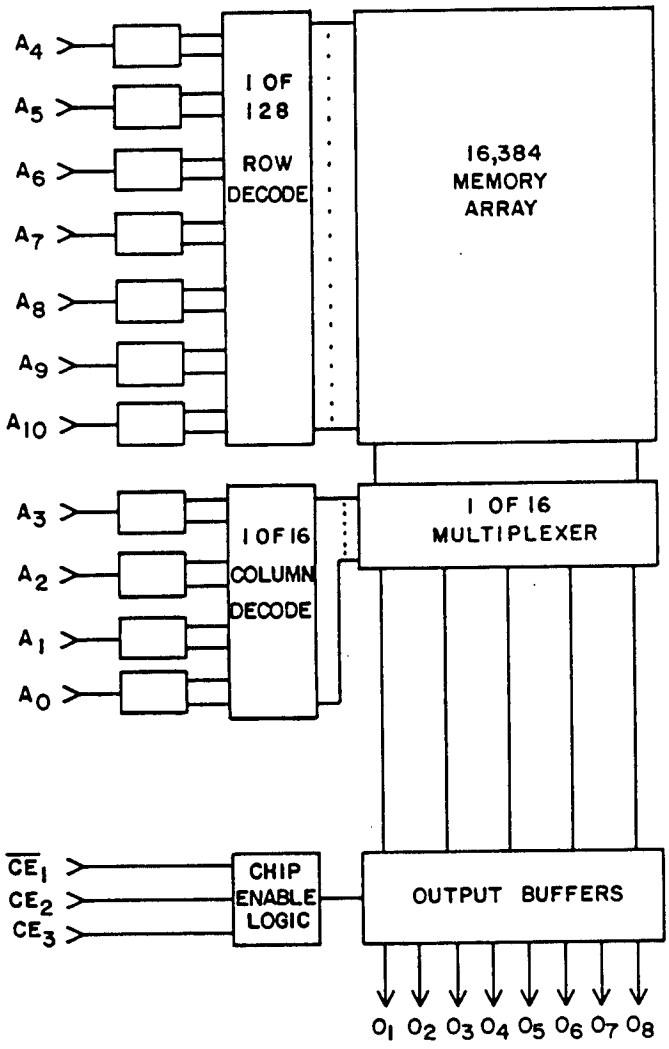


FIGURE 3. Functional block diagrams - Continued.

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Device type 01
Circuit G

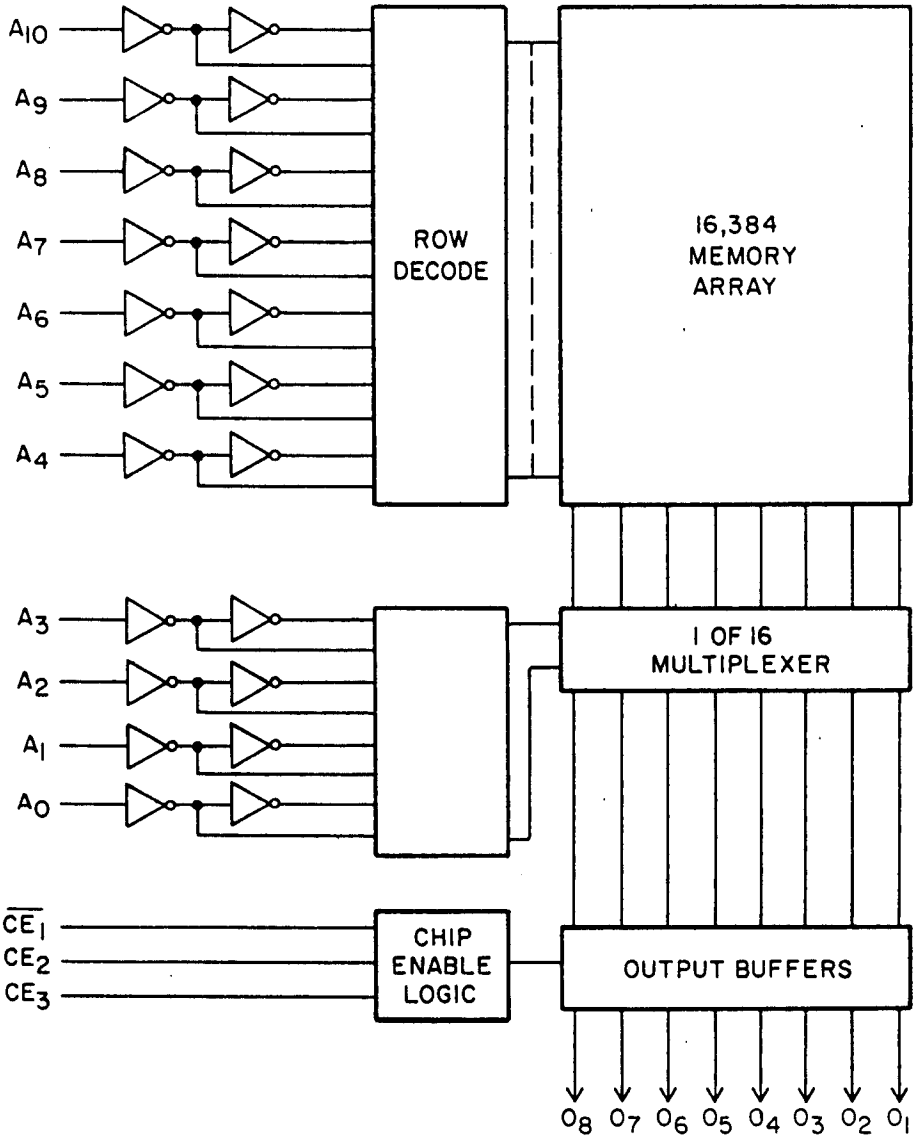


FIGURE 3. Functional block diagrams - Continued.

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Device type 02
Circuit G

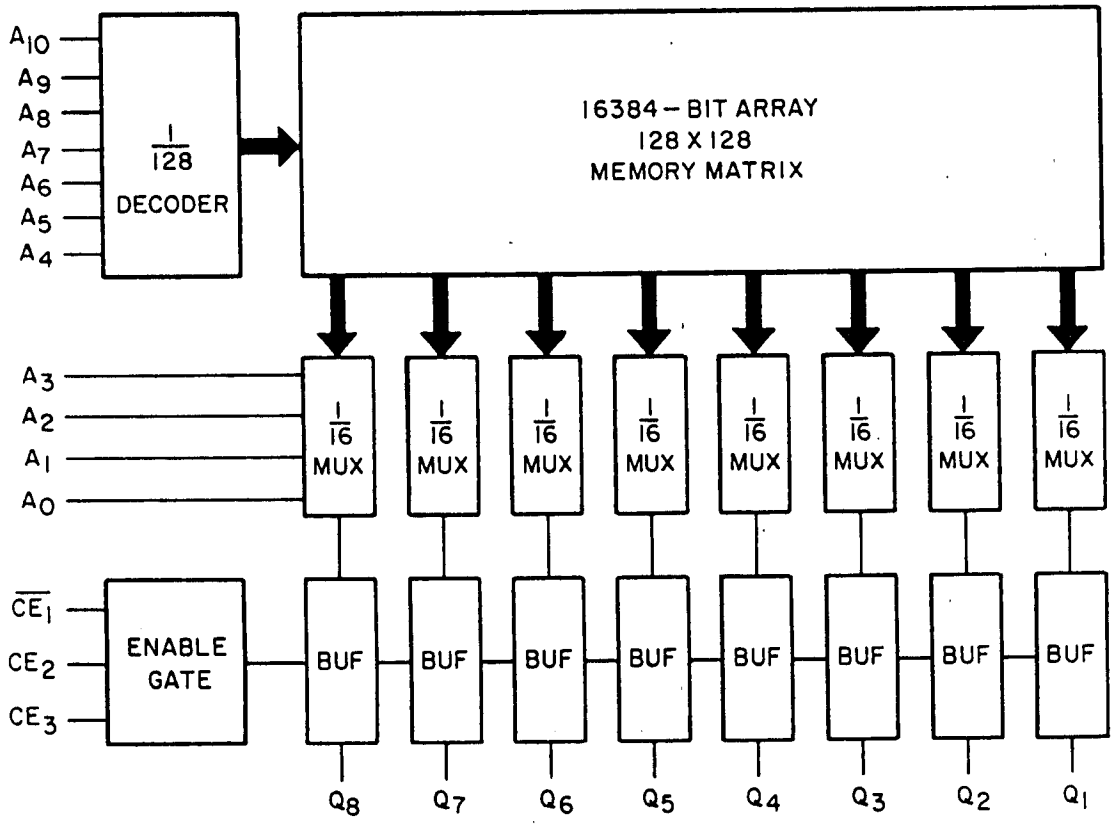


FIGURE 3. Functional block diagrams - Continued.

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Device types 02 and 04
Circuit H

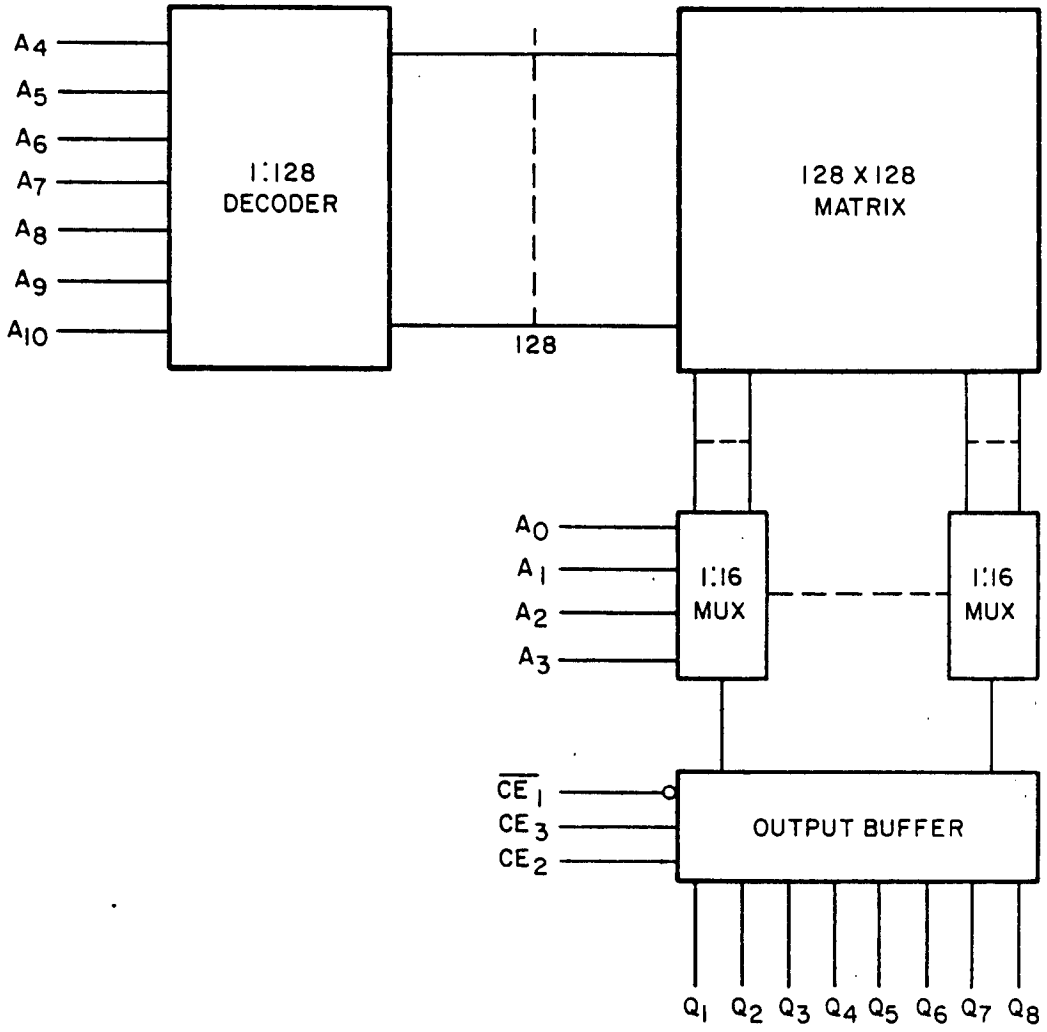


FIGURE 3. Functional block diagrams - Continued.

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Device type 02

Circuit J

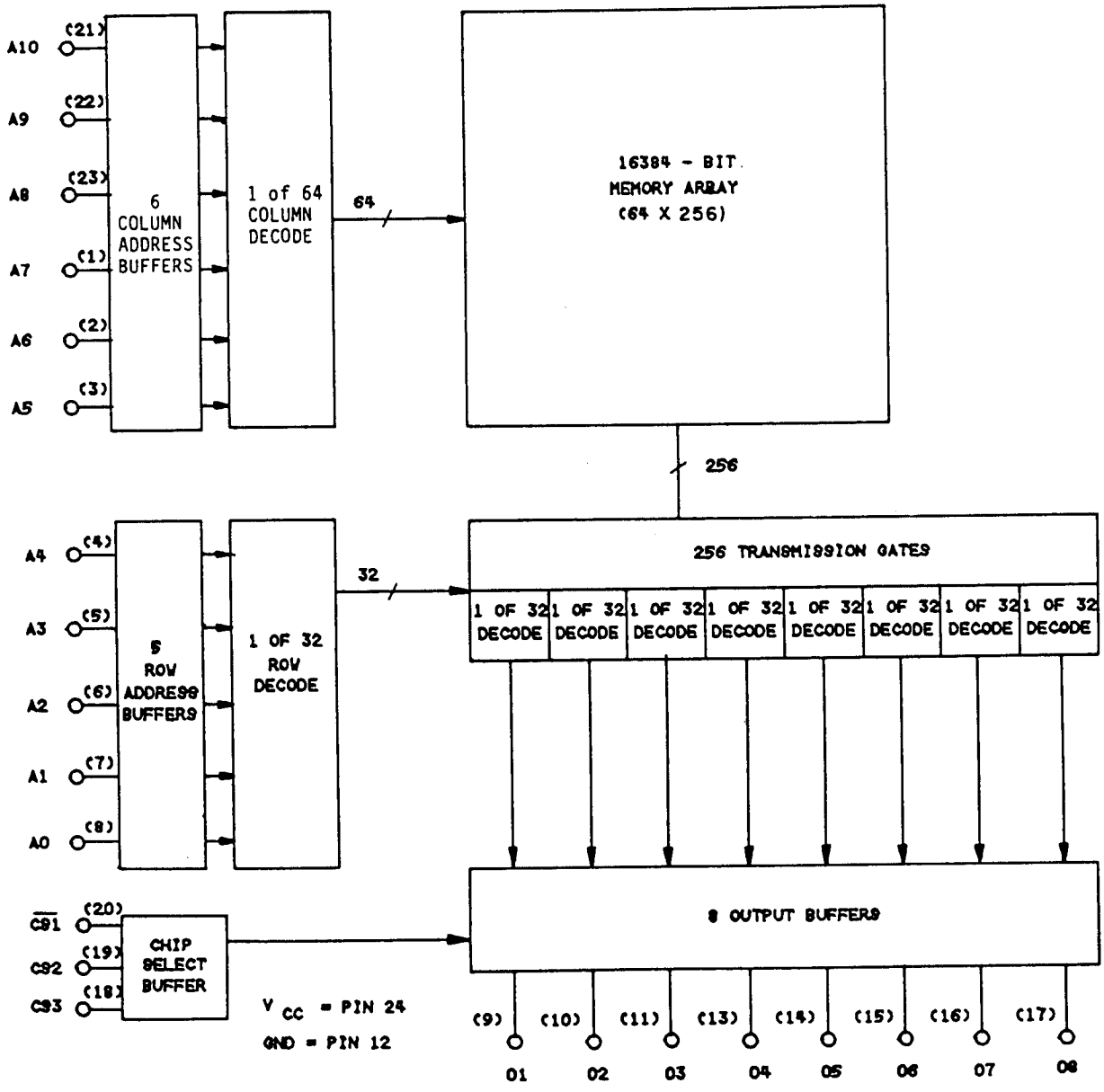
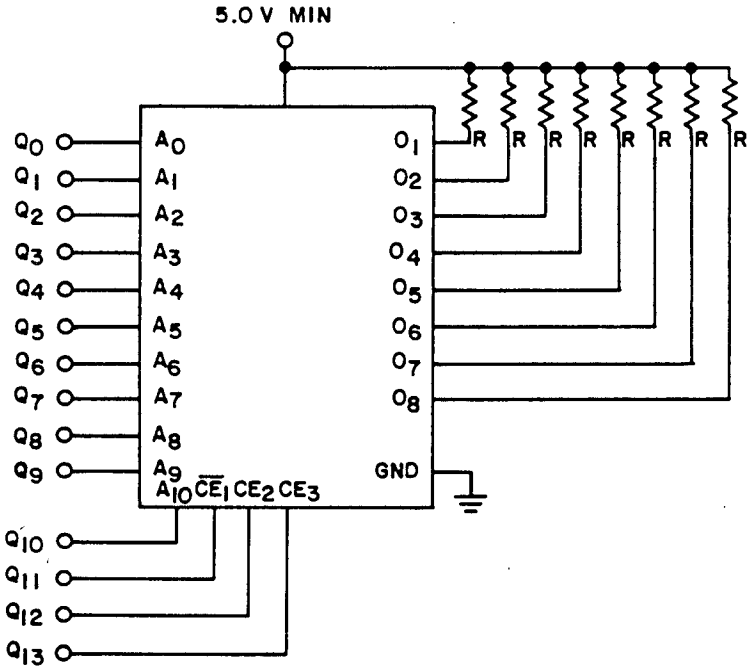


FIGURE 3. Functional block diagrams - Continued.

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Device types 01, 02, 03, and 04



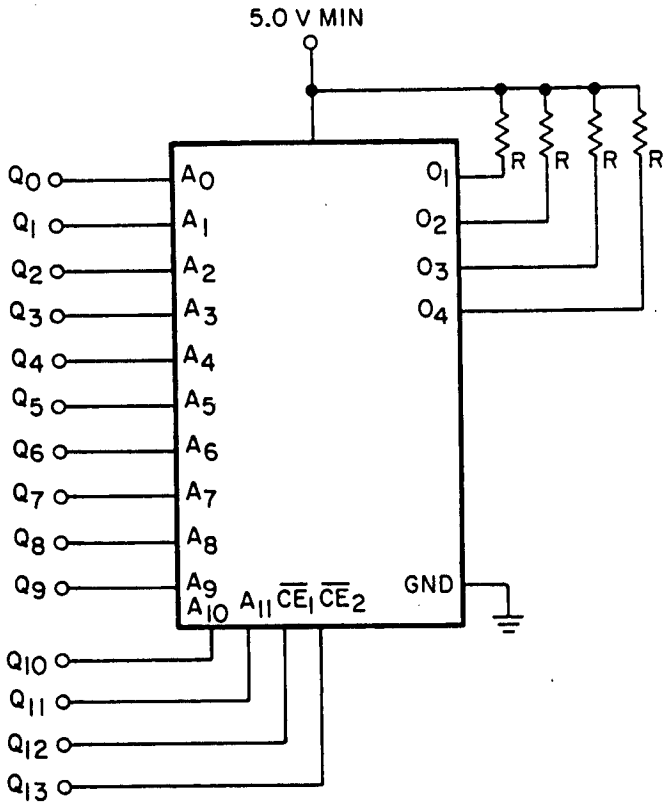
NOTES:
1. R = 560Ω ±5% for circuits C, G and J (300Ω ±5% for circuit A, B, F, and H devices, and 270Ω ±5% for circuit I and circuit D devices). All outputs shall have separate identical loads.
2. All pulse generators have the following characteristics: V_{IL} = -1.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum; 50% ±15% duty cycle and frequencies as specified in note 4.
3. V_{CC} shall be high enough to insure 5.0 V minimum at the device V_{CC} terminal.
4. Input frequencies are as follows:

Input	Frequency (±50%)
Q_0	f_0 = 100 kHz Min
Q_1	f_1 = 1/2 f_0
Q_2	f_2 = 1/2 f_1
Q_3	f_3 = 1/2 f_2
Q_4	f_4 = 1/2 f_3
Q_5	f_5 = 1/2 f_4
Q_6	f_6 = 1/2 f_5
Q_7	f_7 = 1/2 f_6
Q_8	f_8 = 1/2 f_7
Q_9	f_9 = 1/2 f_8
Q_{10}	f_{10} = 1/2 f_9
Q_{11}	f_{11} = 1/2 f_{10}
Q_{12}	f_{12} = 1/2 f_{11}
Q_{13}	f_{13} = 1/2 f_{12}

FIGURE 4. Burn-in and life test circuit.

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Device type 05



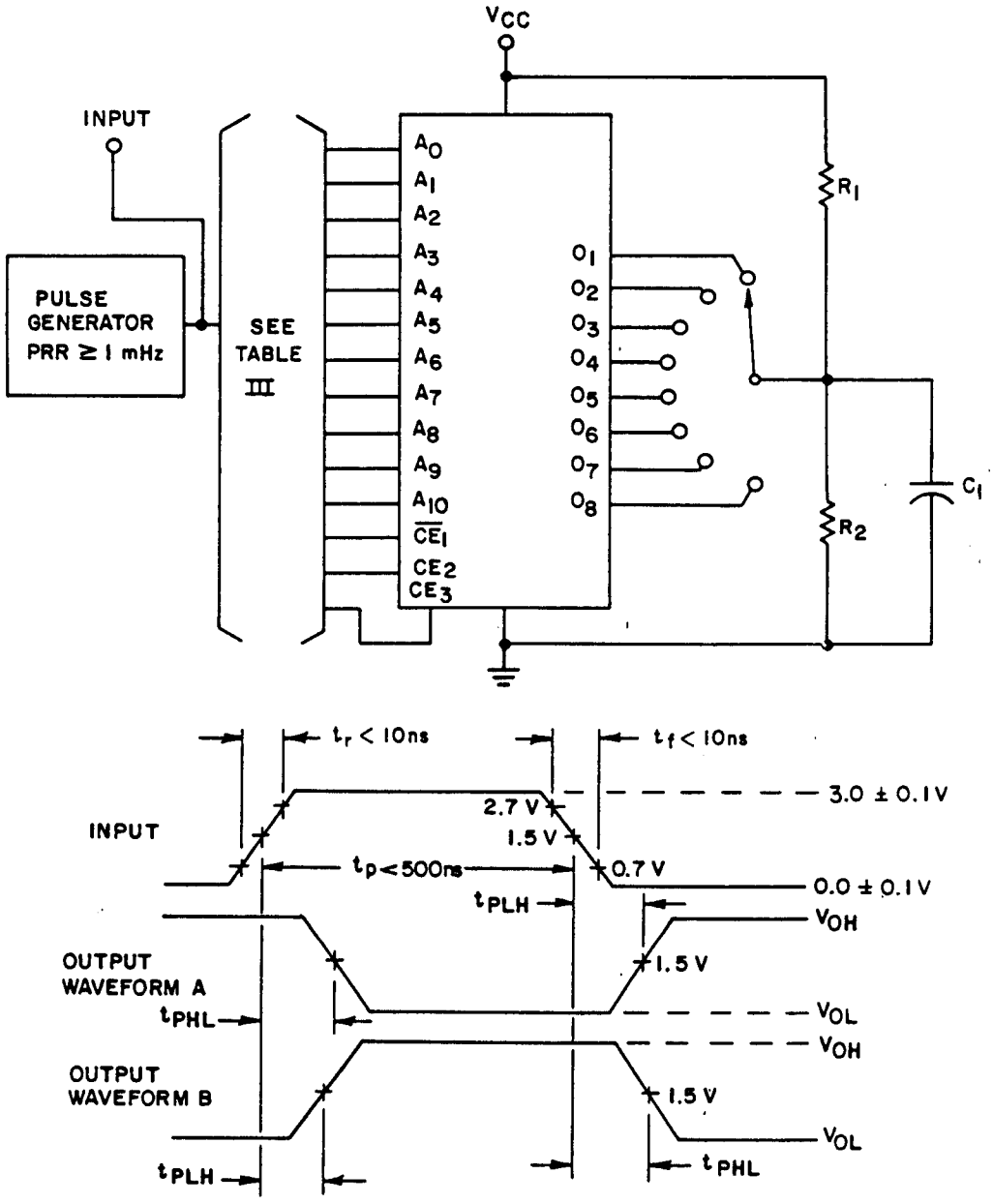
NOTES:
1. R = 300Ω ± 5% for circuits A and C. All outputs shall have separate identical loads.
2. All pulse generators have the following characteristics: V_{IL} = -1.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum; 50% ±15% duty cycle and frequencies as specified in note 4.
3. V_{CC} shall be high enough to insure 5.0 V minimum at the device V_{CC} terminal.
4. Input frequencies are as follows:

Input	Frequency (±50%)
Q_0	f_0 = 100 kHz Min
Q_1	f_1 = 1/2 f_0
Q_2	f_2 = 1/2 f_1
Q_3	f_3 = 1/2 f_2
Q_4	f_4 = 1/2 f_3
Q_5	f_5 = 1/2 f_4
Q_6	f_6 = 1/2 f_5
Q_7	f_7 = 1/2 f_4
Q_8	f_8 = 1/2 f_7
Q_9	f_9 = 1/2 f_8
Q_{10}	f_{10} = 1/2 f_9
Q_{11}	f_{11} = 1/2 f_{10}
Q_{12}	f_{12} = 1/2 f_{11}
Q_{13}	f_{13} = 1/2 f_{12}

FIGURE 4. Burn-in and life test circuit - Continued.

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Device types 01, 02, 03, and 04

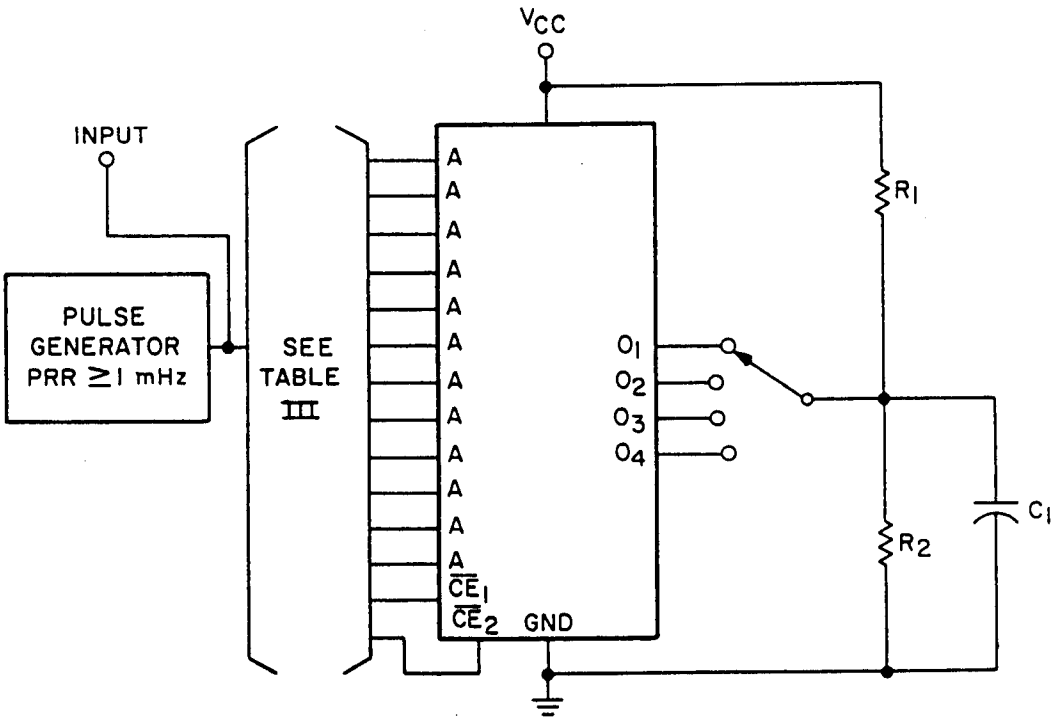


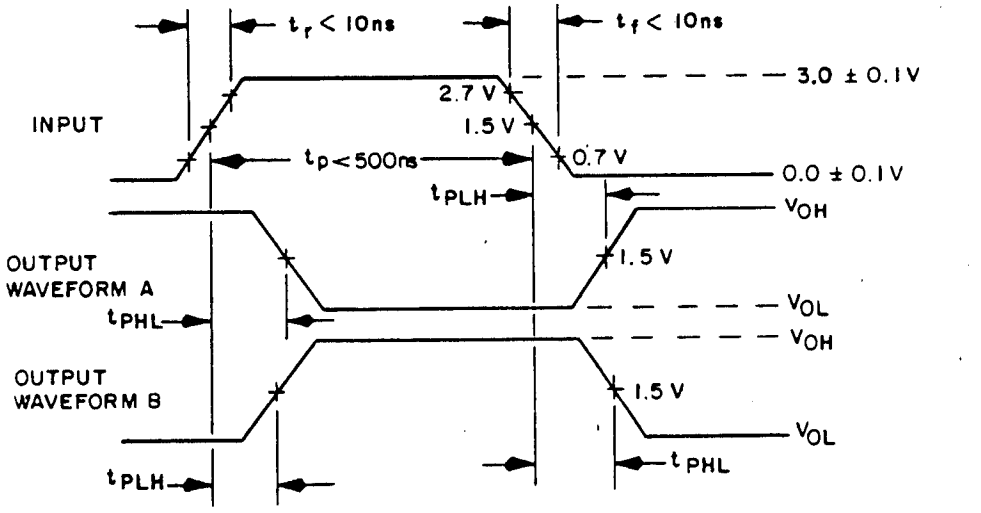
NOTES:
1. Test table for devices programmed in accordance with an altered item drawing may be replaced by the equivalent tests which apply to the specific program configuration for the resulting read-only memory.
2. C_L = 30 pF minimum, including jig and probe capacitance; R_1 = 330Ω ±25% and R_2 = 680Ω ±20%.
3. Outputs may be under load simultaneously.

FIGURE 5. Switching time test circuit.

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Device type 05

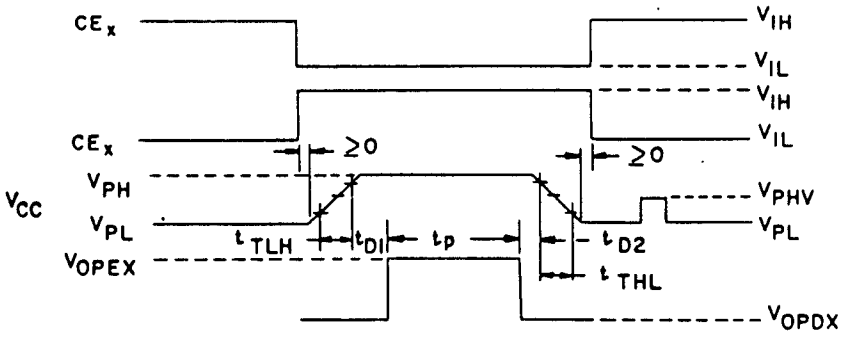




NOTES:
1. Test table for devices programmed in accordance with an altered item drawing may be replaced by the euqivalent tests which apply to the specific program configuration for the resulting read-only memory.
2. C_L = 30 pF minimum, including jig and probe capacitance; R_1 = 330Ω ±25% and R_2 = 680Ω ±20%.
3. Outputs may be under load simultaneously.

FIGURE 5. Switching time test circuit - Continued.

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NOTE:
1. All other waveform characteristics shall be as specified in table IVA.

FIGURE 6A. Programming voltage waveforms during programming for circuit A.

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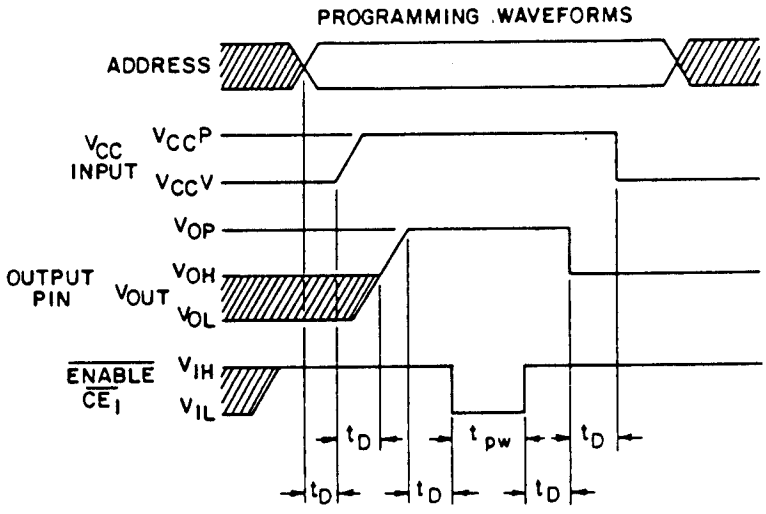
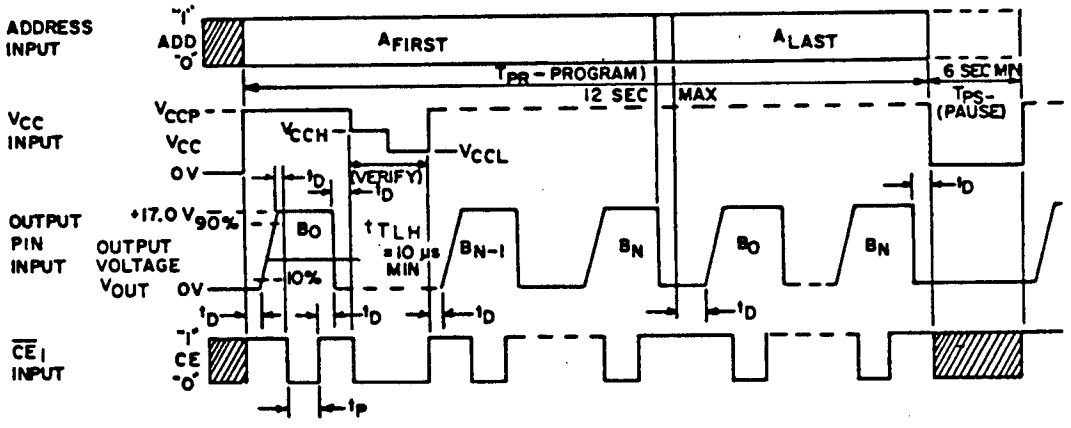


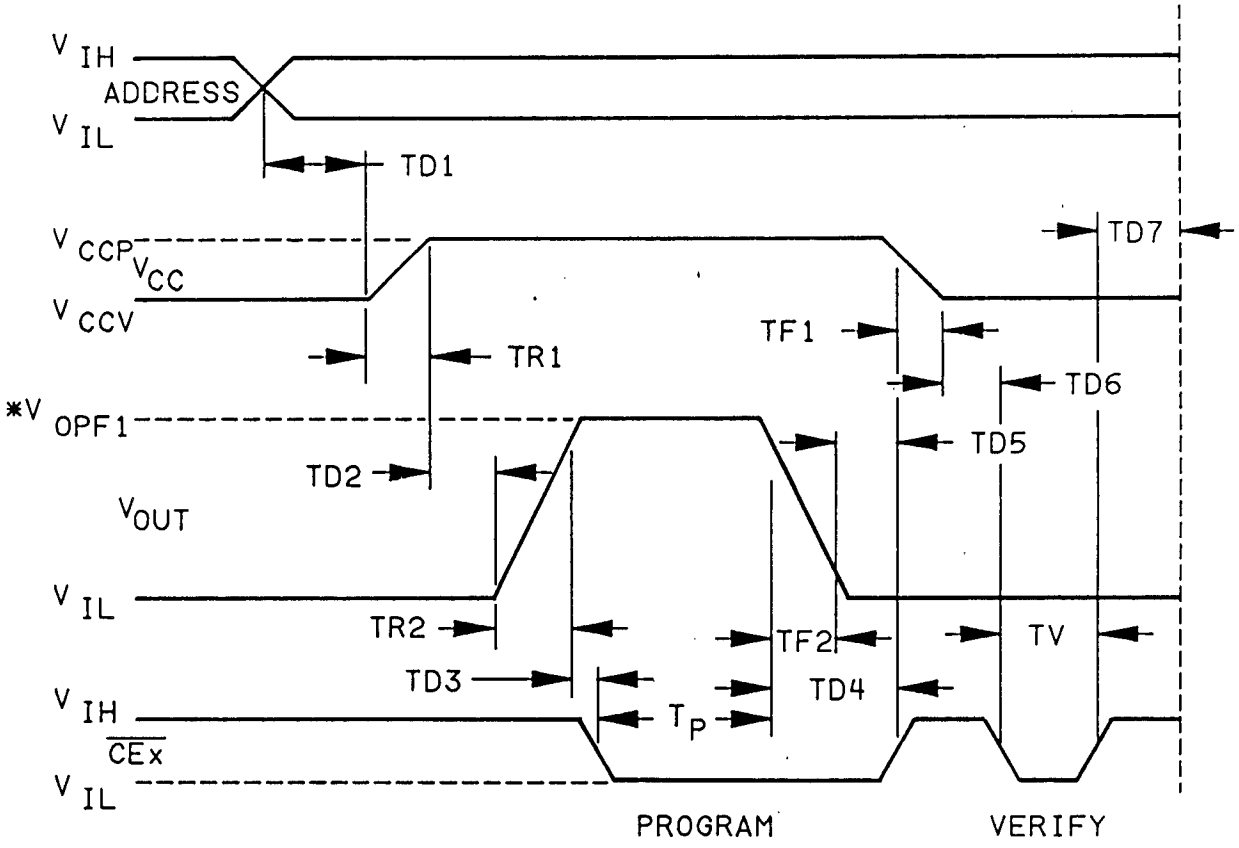
FIGURE 6B. Programming voltage waveforms during programming for circuit B.



NOTE: All other waveform characteristics shall be as specified in table IVC.

FIGURE 6C. Programming voltage waveforms during programming for circuit C, device types 02 and 04.

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*Current clamp or voltage clamp will be needed.

FIGURE 6C. Programming voltage waveforms during programming for circuit C, device type 05 - Continued.

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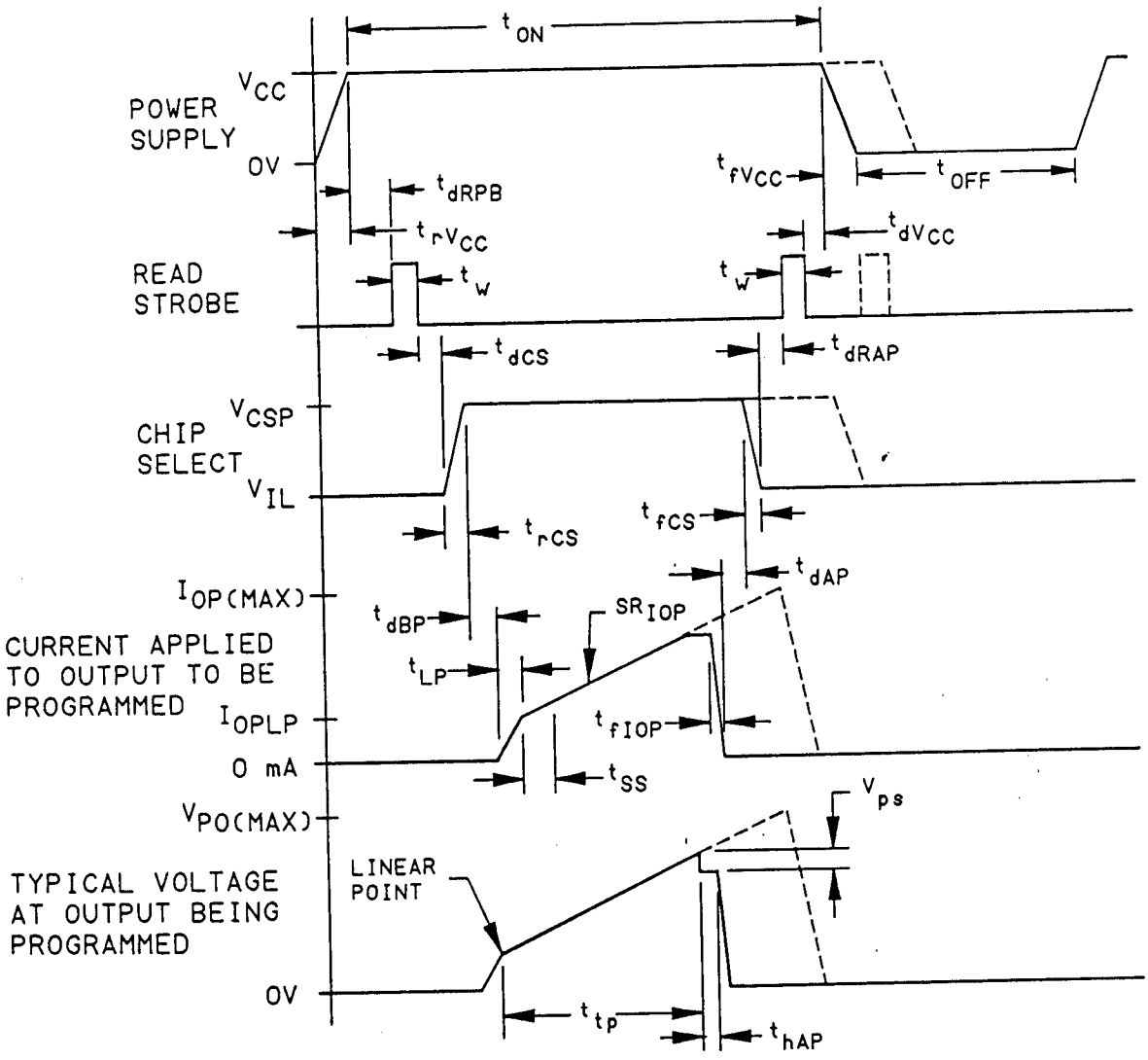
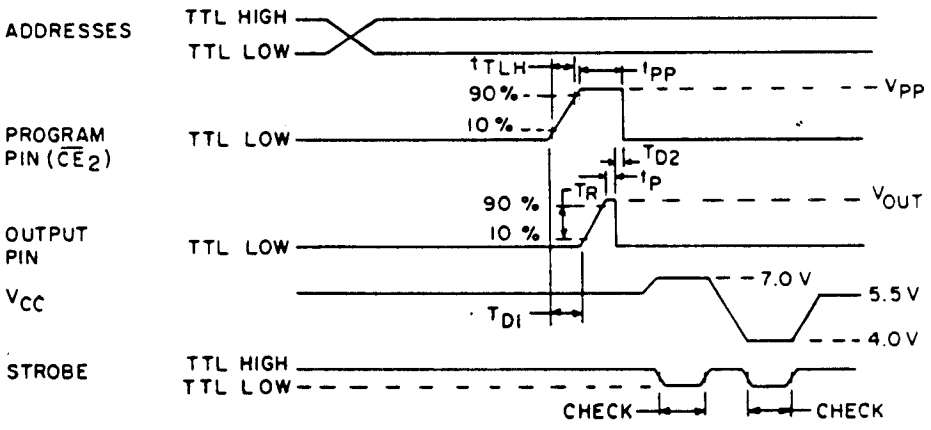


FIGURE 6D. Programming voltage waveforms during programming for circuit D.

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FIGURE 6E. Programming waveforms for circuit E have been discontinued.



NOTES:
1. Output load is 0.2 mA and 12 mA during 7.0 V and 4.0 V check, respectively.
2. All other waveform characteristics shall be as specified in table IVF.

FIGURE 6F. Programming voltage waveforms during programming for circuit F.

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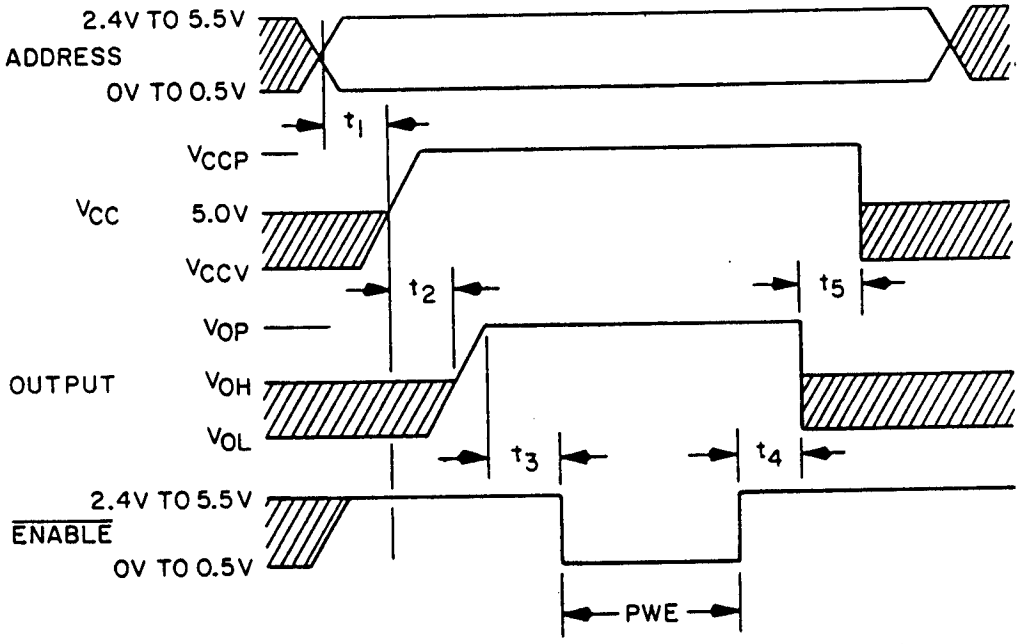


FIGURE 6G. Programming voltage waveforms during programming for circuit G.

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Check the source to verify that this is the current version before use.

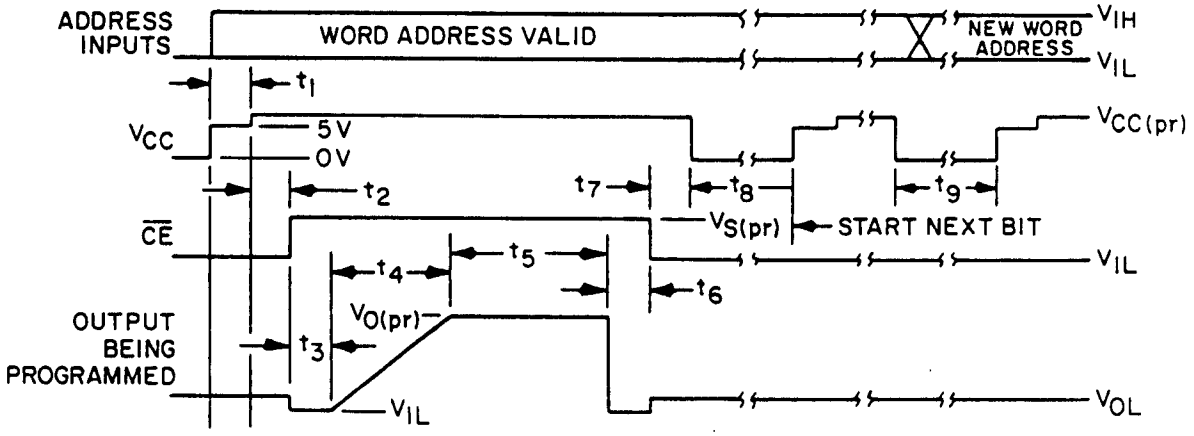
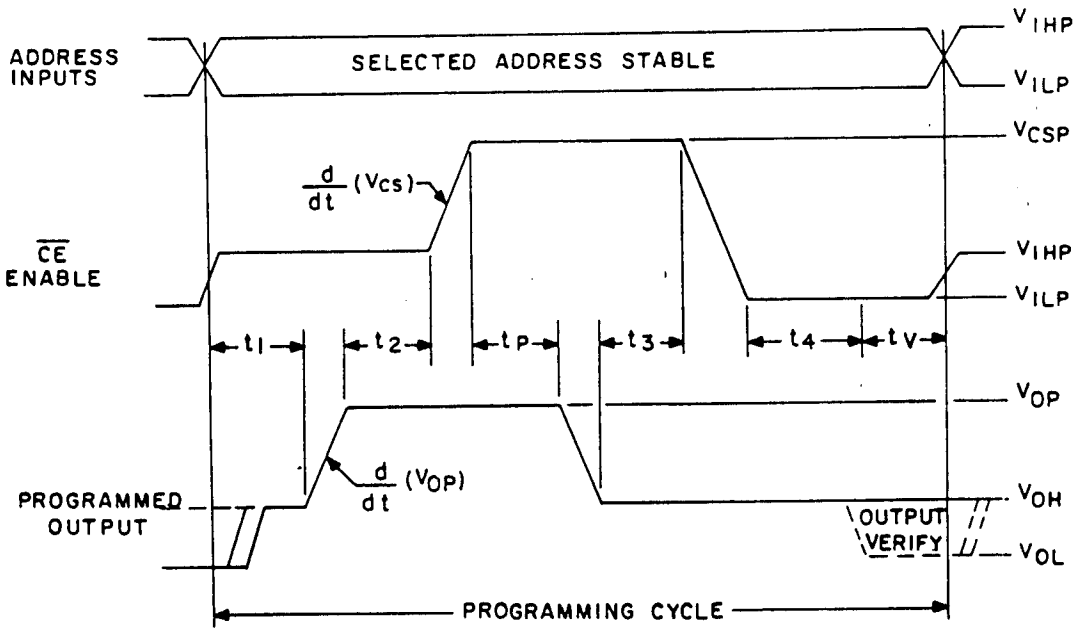


FIGURE 6H. Programming voltage waveforms during programming for circuit H.

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NOTES:
1. All delays between edges are specified from completion of the first edge, not midpoints.
2. Delays t_1, t_2, t_3, and t_4 must be greater than 100 ns; maximum delays of 1 µs are recommended to minimize heating during programming.
3. During t_V the output being programmed is switched to the load R and verified.
4. Outputs not being programmed are connected to V_{ONP} through resistor which provides output current limiting.

FIGURE 6I. Programming voltage waveforms during programming for circuit I.

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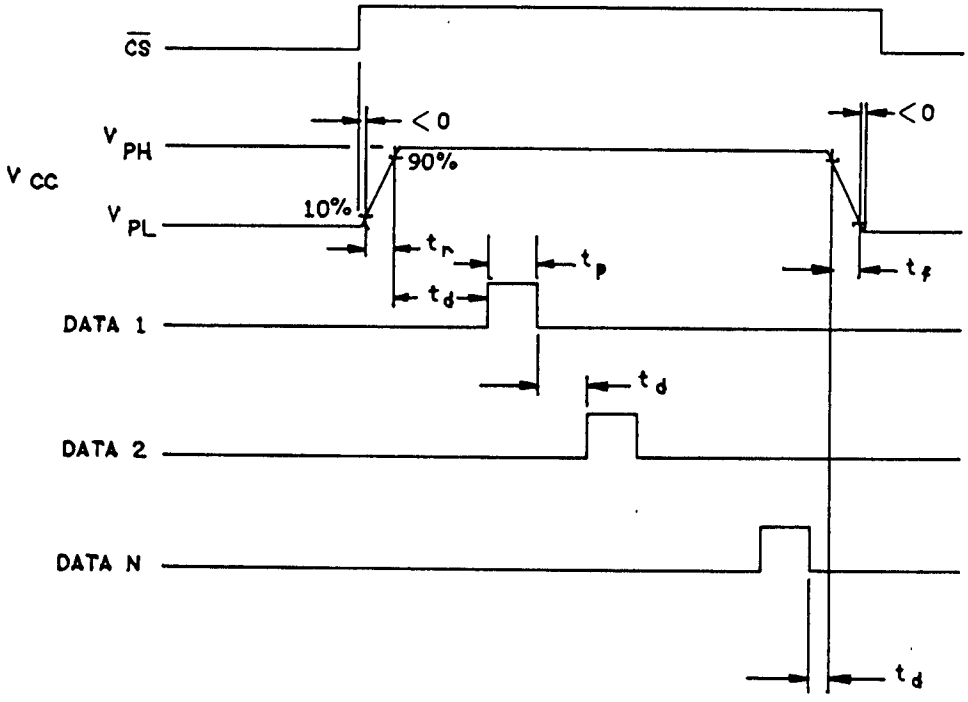
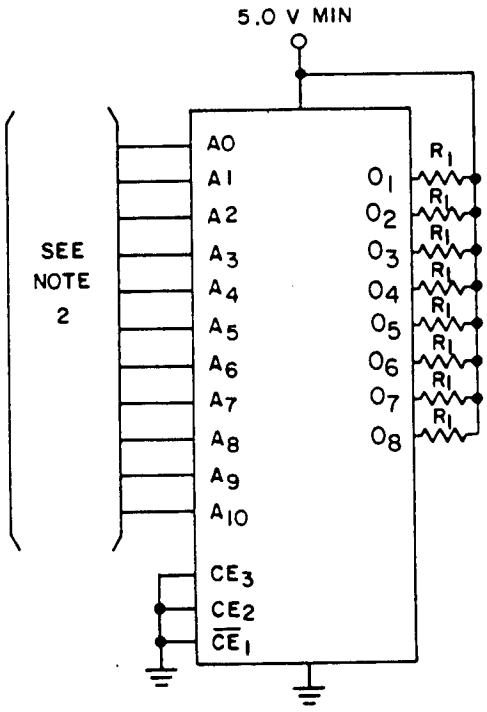


FIGURE 6J. Programming voltage waveforms during programming for circuit J.

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Device types 01, 02, 03, and 04

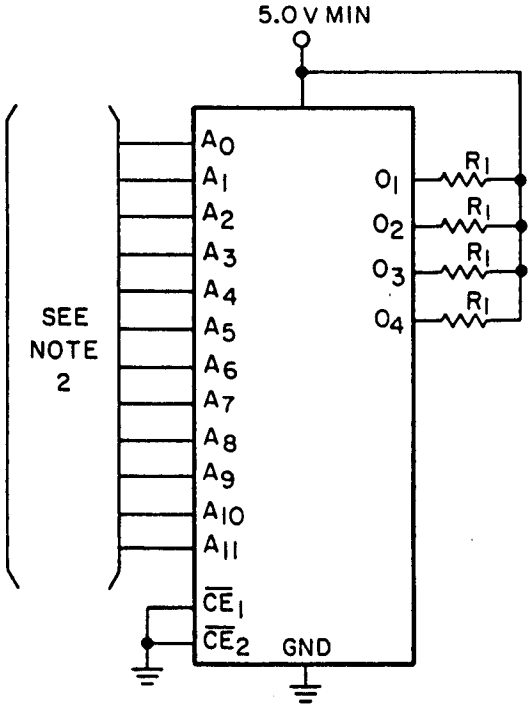


NOTES:
1. R_1 = 4.7 kΩ ±5%. All bit outputs shall have separate identical loads.
2. All address inputs shall be either high, low, or open.
3. Burn-in circuit may be used to perform this test. (See 4.2 d.) All address input shall be either high, low, or open.

FIGURE 7. Freeze-out test bias configuration.

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Device type 05



NOTES:
1. R_1 = 4.7 kΩ ±5%. All bit outputs shall have separate identical loads.
2. All address inputs shall be either high, low, or open.
3. Burn-in circuit may be used to perform this test. (See 4.2 d.) All address input shall be either high, low, or open.

FIGURE 7. Freeze-out test bias configuration - Continued.

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TABLE III. Group A inspection for device types 01 and 03.
(Outputs not designated are open or resistive coupled to GND or voltage.
Terminal conditions: Inputs not designated are high ≥2.0 V or ≤0.8 V.

Subgroup	Symbol	MIL-STD-883-method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	CE_3	CE_2	$\overline{CE_1}$	A_{10}	A_9	A_8	V_{CC}				
1 T_C = +25°C	V_{IC}		1 2 3 4 5 6 7 8 9 10 11 12 13 14	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA				GND " " " " " " " " " " " " "						-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	4.5 V " " " " " " " " " " " " "	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 CE_3 CE_2 $\overline{CE_1}$ A_{10} A_9 A_8		-1.5 " " " " " " " " " " " " "	V " " " " " " " " " " " " "
	V_{OL}	3007 " " " " " "	15 16 17 18 19 20 21 22	1/2/ " " " " " "	1/ 2/ " " " " " "	2/ " " " " " "	2/ " " " " " "	2/ " " " " " "	2/ " " " " " "	2/ " " " " " "	2/ " " " " " "	3/	3/	3/	" " " " " " " " "	3/	3/	3/	3/	3/	2.0 V " " " " " "	2.0 V " " " " " "	0.8 V " " " " " "	2/ " " " " " "	2/ " " " " " "	2/ " " " " " "	" " " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		0.5 " " " " " " "	" " " " " " " "
	I_{IL}	3009 " " " " " " " " " " " " "	23 24 25 26 27 28 29 30 31 32 33 34 35 36	-0.5 V	-0.5 V	-0.5 V	-0.5 V	-0.5 V	-0.5 V	-0.5 V	-0.5 V				" " " " " " " " " " " " " "						-0.5 V	-0.5 V	-0.5 V	-0.5 V	-0.5 V	-0.5 V	5.5 V " " " " " " " " " " " " "	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 CE_3 CE_2 $\overline{CE_1}$ A_{10} A_9 A_8	-1.0 " " " " " " " " " " " " "	-250 " " " " " " " " " " " " "	µA " " " " " " " " " " " " "
	I_{IH}	3010 " " " " " " " " " " " " "	37 38 39 40 41 42 43 44 45 46 47 48 49 50	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				" " " " " " " " " " " " " "						5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	" " " " " " " " " " " " " "	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 CE_3 CE_2 $\overline{CE_1}$ A_{10} A_9 A_8		50 " " " " " " " " " " " " "	" " " " " " " " " " " " " "
	I_{CEX}		51 52 53 54 55 56 57 58									5.2 V	5.2 V	5.2 V	" " " " " " " "	5.2 V	5.2 V	5.2 V	5.2 V	5.2 V	0.5 V	0.5 V	4.5 V				" " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		100 " " " " " " "	" " " " " " " "
	I_{CC}	3005	59	GND	GND	GND	GND	GND	GND	GND	GND				"								GND	GND	GND	GND	"	V_{CC}		185	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 03 - Continued.

Terminal conditions: (Outputs not designated are open or resistive coupled to GND or voltage. Inputs not designated are high ≥2.0 V or ≤0.8 V.

Subgroup	Symbol	MIL-STD-883-method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE}_3$	$\overline{CE}_2$	$\overline{CE}_1$	A_{10}	A_9	A_8	V_{CC}		Min	Max	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C																														
7 T_C = +25°C	Functional tests	4/	60	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																														
9 T_C = +25°C	t_{PHL1} t_{PLH1} t_{PHL2} t_{PLH2}	GALPAT Fig. 5 Sequential Fig. 5	61 62 63 64	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	6/ " " "	6/ " " "	6/ " " "	GND " " "	6/ " " "	6/ " " "	6/ " " "	6/ " " "	6/ " " "	5.5 V 5.5 V 8/ 8/	5.5 V 5.5 V 8/ 8/	GND GND 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	Outputs " " "		7/ 7/ 9/ 9/	ns " " "
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																														

See footnotes at end of table.

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TABLE III. Group A inspection for device types 02 and 04.

Terminal conditions: (Outputs not designated are open or resistive coupled to GND or voltage. Inputs not designated are high ≥2.0 V or ≤0.8 V.

Subgroup	Symbol	MIL-STD-883-method	Case 3	2	3	4	5	6	7	8	9	11	12	13	14	16	17	18	19	20	22	23	24	25	26	27	28	Measured terminal	Test limits		Unit
			Cases J,K,L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Min	Max	
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE_3}$	$\overline{CE_2}$	$\overline{CE_1}$	A_{10}	A_9	A_8	V_{CC}				
1 T_C = +25°C	V_{IC}		1 2 3 4 5 6 7 8 9 10 11 12 13 14	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA				GND						-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	4.5 V	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE_3}$ $\overline{CE_2}$ $\overline{CE_1}$ A_{10} A_9 A_8		-1.5	V
	V_{OL}	3007	15 16 17 18 19 20 21 22	1/2/	1/2/	2/	2/13/	2/	2/	2/	2/	3/	3/	3/		3/	3/	3/	3/	3/	2.0 V	2.0 V	0.8 V	2/13/	2/	2/13/		O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		0.5	
	V_{OH}	3006	23 24 25 26 27 28 29 30	2/14/15/16/17/	2/12/15/16/17/	2/15/16/17/	2/15/16/17/	2/15/16/17/	2/15/16/17/	2/15/16/17/	2/15/16/17/18/	-2 mA	-2 mA	-2 mA		-2 mA	-2 mA	-2 mA	-2 mA	-2 mA				2/15/16/17/	2/15/16/17/	2/15/16/17/		O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8	2.4		
	I_{IL}	3009	31 32 33 34 35 36 37 38 39 40 41 42 43 44	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V										0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	5.5 V	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE_3}$ $\overline{CE_2}$ $\overline{CE_1}$ A_{10} A_9 A_8	-1.0	-250	μA
	I_{IH}	3010	45 46 47 48 49 50 51 52 53 54 55 56 57 58	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V										5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE_3}$ $\overline{CE_2}$ $\overline{CE_1}$ A_{10} A_9 A_8		50	

See footnotes at end of table.

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TABLE III. Group A inspection for device types 02 and 04 - Continued.

Terminal conditions: (Outputs not designated are open or resistive coupled to GND or voltage. Inputs not designated are high ≥2.0 V or ≤0.8 V.

Subgroup	Symbol	MIL-STD-883-method	Case 3	2	3	4	5	6	7	8	9	11	12	13	14	16	17	18	19	20	22	23	24	25	26	27	28	Measured terminal	Test limits		Unit
			Cases J,K,L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Min	Max	
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	CE_3	CE_2	$\overline{CE}_1$	A_{10}	A_9	A_8	V_{CC}				
1 T_C = +25°C	I_{OHZ}		59 60 61 62 63 64 65 66									5.2 V	5.2 V	5.2 V	GND " " " " " " "	5.2 V	5.2 V	5.2 V	5.2 V	5.2 V	0.5 V " " " " " " "	0.5 V " " " " " " "	4.5 V " " " " " " "				5.5 V " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		100 " " " " " " "	µA " " " " " " "
	I_{OLZ}		67 68 69 70 71 72 73 74									0.5 V	0.5 V	0.5 V	" " " " " " " "	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	" " " " " " " "	" " " " " " " "	" " " " " " " "				" " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		-100 " " " " " " "	" " " " " " " "
	I_{OS}	3011 " " " " " "	75 76 77 78 79 80 81 82	14/2/15/16/17/ " " " " " " "	2/15/12/16/17/ " " " " " " "	2/15/16/17/ " " " " " " "	2/15/16/17/ " " " " " " "	2/15/16/17/ " " " " " " "	17/2/15/16/ " " " " " " "	2/15/16/17/ " " " " " " "	18/2/15/16/17/ " " " " " " "	GND	GND	GND	" " " " " " " "	GND	GND	GND	GND	GND	4.5 V " " " " " " "	4.5 V " " " " " " "	0.5 V " " " " " " "	2/15/16/17/ " " " " " " "	2/15/16/17/ " " " " " " "	2/15/16/17/ " " " " " " "	" " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8	-10 " " " " " " "	-100 " " " " " " "	mA " " " " " " "
	I_{CC}	3005	83	GND	GND	GND	GND	GND	GND	GND	GND				"								GND	GND	GND	GND	"	V_{CC}		185	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C																														
7 T_C = +25°C	Functional tests	4/	84	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																														
9 T_C = +25°C	t_{PHL1} t_{PLH1} t_{PHL2} t_{PLH2}	GALPAT Fig. 5 Sequential Fig. 5	85 86 87 88	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	6/ " " "	6/ " " "	6/ " " "	GND " " "	6/ " " "	6/ " " "	6/ " " "	6/ " " "	6/ " " "	5.5 V 5.5 V 8/ 8/	5.5 V 5.5 V 8/ 8/	GND GND 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	Outputs " " "		7/ 7/ 9/ 9/	ns " "
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																														

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05.

Terminal conditions: (Outputs not designated are open or resistive coupled to GND or voltage. Inputs not designated are high ≥2.0 V or ≤0.8 V.

Subgroup	Symbol	MIL-STD-883 method	Case R Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	GND	O_4	O_3	O_2	O_1	$\overline{CE}_2$	$\overline{CE}_1$	A_{11}	A_{10}	A_9	V_{CC}				
1 T_C = +25°C	V_{IC}		1	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	GND					-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	4.5 V	A_8		-1.5	V
			2										"										"	A_7		"	"
			3										"										"	A_6		"	"
			4										"										"	A_5		"	"
			5										"										"	A_4		"	"
			6										"										"	A_3		"	"
			7										"										"	A_2		"	"
			8										"										"	A_1		"	"
			9										"										"	A_0		"	"
			10										"										"	$\overline{CE}_2$		"	"
			11										"										"	$\overline{CE}_1$		"	"
			12										"										"	A_{11}		"	"
			13										"										"	A_{10}		"	"
			14										"										"	A_9		"	"
	V_{OL}	3007	15	1/2/	2/	2/19/	2/19/	2/19/	2/19/	2/19/	2/19/	2/19/	"	3/	3/	3/	3/	0.8 V	0.8 V	2/	2/	2/	"	O_4		0.5	"
		"	16	"	"	"	"	"	"	"	"	"	"					"	"	"	"	"	"	O_3		"	"
		"	17	"	"	"	"	"	"	"	"	"	"					"	"	"	"	"	"	O_2		"	"
		"	18	"	"	"	"	"	"	"	"	"	"					"	"	"	"	"	"	O_1		"	"
	V_{OH}	3006	19	2/	"	2/	2/	2/	2/	2/	2/	2/	"	-2 mA	-2 mA	-2 mA	-2 mA	"	"	"	"	"	"	O_4	2.4		"
		"	20	"	"	"	"	"	"	"	"	"	"					"	"	"	"	"	"	O_3	"		"
		"	21	"	"	"	"	"	"	"	"	"	"					"	"	"	"	"	"	O_2	"		"
		"	22	"	"	"	"	"	"	"	"	"	"					"	"	"	"	"	"	O_1	"		"
	I_{IL}	3009	23	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"					0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	5.5 V	A_8	-1.0	-250	μA
		"	24										"										"	A_7	"	"	"
		"	25										"										"	A_6	"	"	"
		"	26										"										"	A_5	"	"	"
		"	27										"										"	A_4	"	"	"
		"	28										"										"	A_3	"	"	"
		"	29										"										"	A_2	"	"	"
		"	30										"										"	A_1	"	"	"
		"	31										"										"	A_0	"	"	"
		"	32										"										"	$\overline{CE}_2$	"	"	"
		"	33										"										"	$\overline{CE}_1$	"	"	"
		"	34										"										"	A_{11}	"	"	"
		"	35										"										"	A_{10}	"	"	"
		"	36										"										"	A_9	"	"	"
	I_{IH}	3010	37	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"					5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A_8		50	"
		"	38										"										"	A_7		"	"
		"	39										"										"	A_6		"	"
		"	40										"										"	A_5		"	"
		"	41										"										"	A_4		"	"
		"	42										"										"	A_3		"	"
		"	43										"										"	A_2		"	"
		"	44										"										"	A_1		"	"
		"	45										"										"	A_0		"	"
		"	46										"										"	$\overline{CE}_2$		"	"
		"	47										"										"	$\overline{CE}_1$		"	"
		"	48										"										"	A_{11}		"	"
		"	49										"										"	A_{10}		"	"
		"	50										"										"	A_9		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.

Terminal conditions: (Outputs not designated are open or resistive coupled to GND or voltage. Inputs not designated are high ≥2.0 V or ≤0.8 V.

Subgroup	Symbol	MIL-STD-883 method	Case R Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	GND	O_4	O_3	O_2	O_1	$\overline{CE}_2$	$\overline{CE}_1$	A_{11}	A_{10}	A_9	V_{CC}				
1 T_C = +25°C	I_{OHZ}		51										GND	5.2 V				4.5 V	4.5 V				5.5 V	O_4		100	µA
			52										"		5.2 V			"	"				"	O_3		"	"
			53										"			5.2 V		"	"				"	O_2		"	"
			54										"				5.2 V	"	"				"	O_1		"	"
	I_{OLZ}		55										"	0.5 V				"	"				"	O_4		-100	"
			56										"		0.5 V			"	"				"	O_3		"	"
			57										"			0.5 V		"	"				"	O_2		"	"
			58										"				0.5 V	"	"				"	O_1		"	"
	I_{OS}	3011	59	2/	2/	2/	2/	2/	2/	2/	2/	2/	"	GND				0.5 V	0.5 V	2/	2/	2/	"	O_4	-10	-100	mA
		"	60	"	"	"	"	"	"	"	"	"	"		GND			"	"	"	"	"	"	O_3	"	"	"
		"	61	"	"	"	"	"	"	"	"	"	"			GND		"	"	"	"	"	"	O_2	"	"	"
		"	62	"	"	"	"	"	"	"	"	"	"				GND	"	"	"	"	"	"	O_1	"	"	"
	I_{CC}	3005	63	GND	GND	GND	GND	GND	GND	GND	GND	GND	"					GND	GND	GND	GND	GND	"	V_{CC}		185	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C.																										
7 T_C = +25°C	Functional tests	4/	64	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PHL1} t_{PLH1} t_{PHL2} t_{PLH2}	GALPAT Fig.5 Sequential Fig. 5	65 66 67 68	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	GND " " "	6/ " " "	6/ " " "	6/ " " "	6/ " " "	GND GND 8/ 8/	GND GND 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	5/ 5/ 8/ 8/	Outputs " " "		80 80 40 40	ns " " "
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																										
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																										

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1/ For unprogrammed devices, apply 13.0 V on pin 1(A_7) and pin 2(A_6), for device types 01 and 02, and on pin 1 (A_8) for device type 05 for circuit A devices.

2/ For programmed devices, select an appropriate address to acquire the desired output state. V_{IH} = 2.0 V, V_{IL} = 0.8 V.

3/ I_{OL} = 8 mA for circuits C and G.

I_{OL} = 16 mA for circuits A, B, D, F, H, I, and J.

4/ The functional tests shall verify that no fuses are blown for unprogrammed devices or that the altered item drawing pattern exists for programmed devices (see table II and 3.2.2.2). All bits shall be tested. Terminal conditions shall be as follows:

a. Inputs: H = 2.4 V, L = 0.4 V

b. Outputs: Output voltage shall be:

$H \geq 1.5$ V and $L \leq 1.5$ V.

c. The functional tests shall be performed with V_{CC} = 4.5 V and V_{CC} = 5.5 V.

5/ GALPAT (programmed PROM). This program will test all bits in the array, the addressing and interaction between bits for ac performance, t_{PHL1}, t_{PLH1}. Each bit in the pattern is fixed by being programmed with an "H" or "L".

Description:

1. Word 0 is read.
2. Word 1 is read.
3. Word 0 is read.
4. Word 2 is read.
5. Word 0 is read.
6. The reading procedure continues back and forth between word 0 and the next higher numbered word until word 2047 or 4095 is reached, then increments to the next word and reads back and forth as in steps 1 through 7 and shall include all words.
7. Pass execution time = $(n^2 + n)$ x cycle time, n = 2048 or 4096.
8. The GALPAT tests shall be performed with V_{CC} = 4.5 V and 5.5 V.

6/ The outputs are loaded per figure 5.

7/ t_{PHL1}, t_{PLH1} = 100 ns for device types 01 and 02 and 55 ns for device types 03 and 04.

8/ Sequential test (programmed PROM). This program will test all bits in the array for t_{PHL2} and t_{PLH2}.

Description:

1. Each word in the pattern is tested from the enable lines to the output lines for recovery.
2. Word 0 is addressed. Enable line is pulled HI to LO and LO to HI. t_{PHL2} and t_{PLH2} are read.
3. Word 1 is addressed. Same enable sequence as above.
4. The reading procedure continues until word 2047 or 4095 is reached.
5. Pass execution time = 2048 x cycle time (or 4096 x cycle time).
6. The sequential tests shall be performed with V_{CC} = 4.5 V and 5.5 V.

9/ t_{PHL2}, t_{PLH2} = 50 ns for device types 01 and 02 and 30 ns for device types 03 and 04.

10/ For unprogrammed devices, apply 13 V on pin 8(A_0) for circuit I devices.

11/ For unprogrammed devices, 12.0 V on pin 6(A_2) and 0.0 V on pin 5(A_3) for circuit F devices.

12/ For unprogrammed devices, apply 13 V on pin 2(A_6) for circuit I devices.

13/ For unprogrammed devices, apply 10 V to pin 4(A_4), apply V_{OH} to pin 21 (A_{10}), and apply V_{OL} to pin 23(A_8) for circuit H.

14/ For unprogrammed devices, apply 10.5 V on pin 1(A_7) for circuit B devices.

15/ For unprogrammed devices, apply 10.5 V to pin 3(A_5), apply 0 V to pins 4, 5, 6, 7, 8 (A_4, A_3, A_2, A_1, A_0), and apply 3 V to pins 1, 2, 21, 22, 23, (A_7, A_6, A_{10}, A_9, A_8) for circuit G devices.

16/ For unprogrammed device, type 02 (82S191), with date codes before 8626, apply 10.0 V on pin 6(A_2); apply 5.0 V to all other addresses for circuit C devices.

17/ For unprogrammed device types 02 (with date codes of 8626 or later) and 04 (82S191A), apply 10.0 V on A_4; apply 5.0 V on A_0, A_1, A_2, A_3 and A_6; and apply 0.5 V on A_5, A_7, A_8, A_9 and A_{10} for circuit C devices.

18/ For unprogrammed devices, apply 12.0 V on pin 8(A_0) for circuit D devices.

19/ For unprogrammed device type 05, apply 15.0 V to pin 4(A_5); apply 0.0 V to pins 5, 9(A_4, A_0); apply 4.5 V to pins 3, 6, 7, 8(A_6, A_3, A_2, A_1) for circuit C devices.

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TABLE IVA. Programming characteristics for circuit A.

Parameter	Symbol	Limits 1/			Unit
		Min	Recommended	Max	
Address input voltage 2/	V_{IH} V_{IL}	2.4 0.0	5.0 0.4	5.0 0.5	V
Programming Voltage to V_{CC} low Program verify Verify voltage	V_{PH} 3/ V_{PL} V_{PHV} V_R 4/	10.75 0.0 --- 4.5	11.0 0.0 5.5 ---	11.25 1.5 --- 5.5	V " " "
Programming input low current at V_{PH}	I_{ILP}	---	-300	-600	µA
Programming voltage (V_{CC}) transition time	t_{TLH} t_{THL}	1 1	5 5	10 10	µs µs
Programming delay	t_{D1} t_{D2}	10 1	10 5	20 5	µs
Programming pulse width	t_p 5/	90	100	110	µs
Programming duty cycle	PDC	---	30	60	%
Output voltage Enable Disable	V_{OPE} 6/ V_{OPD}	10.5 0.0	10.5 5.0	11.0 5.5	V V

During the programming the chip must be disabled for proper operation.

1/ T_C = 25°C.

2/ No inputs should be left open for V_{IH}.

3/ V_{PH} source must be capable of supplying one ampere.

4/ It is recommended that post programming dual verification be made at V_R min and V_R max.

5/ Note step j in programming procedure.

6/ V_{OPE} source must be capable of supplying 10 mA minimum.

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TABLE IVB. Programming characteristics for circuit B.

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
V_{CC} required during programming	V_{CCP}		10.5	11.0	11.5	V
V_{OUT} current limit during programming	I_{OP}		20	25	30	mA
Output programming voltage	V_{OUT}		10.5	11.0	11.5	V
Pulse width of programming voltage	t_p		9	10	11	µs
Programming delay	t_D		0	1	10	µs
V_{CCP} or V_{OUT} transition time	t_{TLH}	Rise time of V_{CC} or V_{OUT}	1	5	10	V/µs
V_{CCP} current	I_{CCP}		800	900	1,000	mA
Low V_{CC} for verification	V_{CCL}		3.9	4.0	4.1	V
High V_{CC} for verification	V_{CCH}		5.8	6.0	6.2	V
Address input voltage	V_{IH} V_{IL}		2.4 0.0	5.0 0.4	5.5 0.8	V V
Maximum duty cycle during automatic programming of program pin and output pin	D.C.	t_p/tc	---	25	50	%

1/ T_C = +25°C.

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TABLE IVC. Programming characteristics for circuit C, device types 02 and 04.

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
Programming voltage to V_{CC}	V_{CCP} 2/	I_{CCP} = 375 ±75 mA transient or steady state	8.5	8.75	9.0	V
Verification upper limit	V_{CCH}		5.3	5.5	5.7	V
Verification lower limit	V_{CCL}		4.3	4.5	4.7	V
Verify threshold	V_S 3/		1.4	1.5	1.6	V
Programming supply current	I_{CCP}	V_{CCP} = +8.75 ±.25 V	300		450	mA
Input voltage high level "1"	V_{IH}		2.4		5.5	V
Input voltage low level "0"	V_{IL}		0	0.4	0.8	V
Input current	I_{IH}	V_{IH} = +5.5 V			50	µA
Input current	I_{IL}	V_{IL} = +0.4 V			-500	µA
Output programming voltage	V_{OUT} 4/	I_{OUT} = 200 ±20 mA; transient or steady state	16	17	18	V
Output programming current	I_{OUT}	V_{OUT} = +17 V ±1 V	180	200	220	mA
Programming voltage transition time	t_{TLH}		10		50	µs
$\overline{CE}$ programming pulse width	t_p		300	400	500	µs
Pulse sequence delay	t_D		10			µs

1/ T_C = +25°C.

2/ Bypass V_{CC} to GND with a 0.01 µF capacitor to reduce voltage spikes.

3/ V_S is the sensing threshold of the PROM output voltage for a programmed bit. It normally constitutes the reference voltage applied to a comparator circuit to verify a successful fusing attempt.

4/ Care should be taken to insure the 17 V ±1 output voltage is maintained during the entire fusing cycle. The recommended supply is a constant current source clamped at the specified voltage limit.

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TABLE IVC. Programming characteristics for circuit C, device type 05. - Continued

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
Programming voltage to V_{CC} 2/	V_{CCP}	I_{CCP} = 425 ±75 mA transient or steady state	8.5	8.75	9.0	V
Verify voltage	V_{CCV}		4.75	5.0	5.25	V
Input voltage high level "1"	V_{IH}	I_{IH} = 50 µA	2.4	3.0	5.5	V
Input voltage low level "0"	V_{IL}	I_{IL} = 500 µA	0.0	0.0	0.5	V
Forced output current	I_{OPF}	V_{OPF} = 17.5 ±0.5 V	150	185	220	mA
Forced output voltage (program) 3/	V_{OPF1}	I_{OPF} = 300 ±25 mA	17.0	17.5	18.0	V
Forced output voltage (program) 3/	V_{OPF2}	I_{OPF} = 300 ±25 mA	20.0		22.0	V
Output voltage high	V_{OH}		2.4		5.25	V
Output voltage low	V_{OL}		0.0		0.45	V
V_{CC} delay time	T_{D1}	50% to 10% V_{CCP}	10	10	25	µs
V_{OUT} delay time	T_{D2}	90% V_{CCP} to 10% V_{OFF}	1.0	1.0	5.0	µs
Pulse sequence delays	T_{D3} - T_{D8}	See figure 6C	1.0	1.0	10	µs
V_{CC} rise time	T_{R1}	0% to 100%	4.0	7.0	8.0	µs
V_{OUT} rise time	T_{R2}	10% to 90%	3.0	10	17	µs
V_{CC} fall time	T_{F1}	100% to 0%	2.0	4.0	10	µs

See footnotes at end of table.

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TABLE IVC. Programming characteristics for circuit C, device types 05. - Continued

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
V_{OUT} fall time	T_{F2}	100% to 0%	4.0	7.0	20	µs
$\overline{CE}_2$ programming pulse width 4/	Tp	10% to 10%	5.0	10	30	µs
$\overline{CE}_2$ verify pulse width 4/	Tv	10% to 10%	5.0	5.0	10	µs
Clock pulse width (CK)	T_{WC}	50% to 50%	0.5	0.75	1.0	µs

1/ T_C = +25°C.

2/ If the overall program/verify cycle exceeds the recommended value, a 25% duty cycle must be used for V_{CCP}.

3/ V_{OPF} supply should regulate to ±0.25 V at I_{OPF}. Maximum slew rate for V_{OPF} should be 1.0 V/µs.

4/ $\overline{CE}_2$ rise time slew rate should be 1.0 V/ns maximum. $\overline{CE}_2$ fall time slew rate should be 10.0 V/ns maximum.

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TABLE IVD. Programming characteristics for circuit D.

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
Power supply voltage	V_{CC}		6.4	6.5	6.6	V
Power supply rise time 2/	$t_{r(VCC)}$		0.2	2.0		µs
Power supply fall time 2/	$t_{f(VCC)}$		0.2	2.0		µs
V_{CC} on time 3/	t_{ON}	See programming				
V_{CC} off time 4/	t_{OFF}	Time diagram				
Duty cycle for V_{CC}		$\frac{t_{ON}}{t_{OFF} + t_{ON}}$			50	%
Read delay before programming	t_{dRBP}	Initial check		3.0		µs
Fuse read time	t_W 5/			1.0		µs
Delay to V_{CC} off	$t_{d(VCC)}$ 5/			1.0		µs
Delay to read after programming	t_{dRAP} 5/	Programming verification		3.0		µs
Chip select programming voltage	V_{CSP}		20.0	20.0	22.0	V
Chip select program current limit	I_{CSP}		175	180	185	mA
Input voltage low	V_{IL}		0.0	0.0	0.4	V
Input voltage high	V_{IH}		2.4	5.0	5.0	V
Delay to chip deselect	t_{dCS}			1.0		µs
Chip select pulse rise time	t_{rCS}		3.0	4.0		µs
Delay to chip select time	t_{dAP}		0.2	1.0		µs
Chip select pulse fall time	t_{fCS}		0.1	0.1	1.0	µs

See footnotes at end of table.

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TABLE IVD. Programming characteristics for circuit D - Continued.
Ramp characteristics

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
Programming current linear point	I_{OPLP}			10	20	mA
Output programming current limit	$I_{OP(MAX)}$	Apply current ramp to selected output	155	160	165	mA
Output programming voltage limit	$V_{OP(MAX)}$		24	25	26	V
Current slew rate	SR_{IOP}	Constant after linear point	0.9	1.0	1.1	mA/µs
Blow sense voltage	V_{PS}		0.7			V
Delay to programming ramp	t_{dBP}		2.0	3.0		µs
Time to reach linear point	t_{LP}		0.2	1.0	10	µs
Program sense inhibit	t_{ss}		2.0	3.0	10	µs
Time to program fuse	t_{tp}		3.0		150	µs
Programming ramp hold time	t_{hAP}	After fuse programs	1.4	1.5	1.6	µs
Programming ramp fall time 2/	t_{fIOP}			0.1	0.2	µs

1/ T_C = +25°C.

2/ Rise and fall times are from 10% to 90%.

3/ Total time V_{CC} is on to program fuse is equal to or greater than the sum of all the specified delays, pulse widths and rise/fall times.

4/ t_{OFF} is equal to or greater than t_{ON}.

5/ Proceed to next address after read strobe indicates programmed cell.

TABLE IVE. Programming characteristics for circuit E. - discontinued

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TABLE IVF. Programming characteristics for circuit F.

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
V_{CC} required during programming	V_{CCP}		5.4	5.5	5.6	V
Rise time of program pulse to data out or program pin	t_{TLH}		0.34	0.40	0.46	V/µs
Programming voltage on program pin	V_{PP}		32.5	33	33.5	V
Output programming voltage	V_{OUT}		25.5	26	26.5	V
Programming pin pulse width ($\overline{CE}$)	t_{PP}	Chip disabled V_{CC} = 5.5 V	---	100	180	µs
Pulse width of programming voltage	t_P		1		40	µs
Required current limit of power supply feeding program pin and output during program	I_L	V_{PP} = 33 V, V_{OUT} = 26 V, V_{CC} = 5.5 V	240		---	mA
Required time delay between disabling memory output and application of output programming pulse	T_{D1}	Measured at 10% levels	70	80	90	µs
Required time delay between removal of programming pulse and enabling memory output	T_{D2}		100			ns
Output current during verification	I_{OLV1}	Chip enabled V_{CC} = 4.0 V	11	12	13	mA
	I_{OLV2}	Chip enabled V_{CC} = 7.0 V	0.19	0.2	0.21	mA
Address input voltage	V_{IH}		2.4	5.0	5.5	V
	V_{IL}		0.0	0.4	0.8	V
Maximum duty cycle during automatic programming of program pin and output pin	D.C.	t_p/tc	---	---	25	%

1/ T_C = 25°C.

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TABLE IVG. Programming characteristics for circuit G.

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended value	Max	
Required V_{CC} for programming	V_{CCP}		10.0	10.5	11.0	V
I_{CC} during programming	I_{CCP}	V_{CC} = 11 V			750	mA
Required output voltage for programming	V_{OP}		10.0	10.5	11.0	V
Output current while programming	I_{OP}	V_{OUT} = 11 V			20	mA
Rate of voltage change of V_{CC} or output	I_{RR}		1.0		10.0	V/µs
Programming pulse width (enabled)	PWE		9	10	11	µs
Required V_{CC} for verification	V_{CCV}		3.8	4.0	4.2	V
Maximum duty cycle for V_{CC} at V_{CCP}	MDC			25	25	%
Address set-up time	t_1		100			ns
V_{CCP} set-up time	t_2	2/	5			µs
V_{CCP} hold time	t_5		100			ns
V_{OP} set-up time	t_3		100			ns
V_{OP} hold time	t_4		100			ns

1/ T_C = +25°C.

2/ V_{CCP} setup time may be greater than 0 if V_{CCP} rises at the same rate or faster than V_{OP}.

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TABLE IVH. Programming characteristics for circuit H. 1/

Parameters	Symbol	Min	Nom	Max	Unit
Steady-state supply voltage	V_{CC}	4.75	5	5.25	V
Input voltage	V_{IH}	3	4	5	V
	V_{IL}	0	0	0.5	
Voltage all outputs exept the one to be programmed		0	0	0.5	V
Supply voltage level to program a bit	$V_{CC(pr)}$	5.75	6	6.25	V
Select or enable level to program a bit	$V_{S(pr)}$	9.75	10	11	V
Output level during interval t5	$V_{O(PR)}$	15.75	16	16.25	V
Supply voltage during vertification (see step 0)	Low	4.4	4.5	4.6	V
	High	5.4	5.5	5.6	
Time for V_{CC} to settle and to verify need to program	t_1	0	5	10	µs
Timing from V_{CC} = 6 V until chip select (enable) is at 10 V	t_2	5	5	10	µs
Timing from chip select (enable) high to start of program ramp	t_3	0.1	5	10	µs
Ramp time, output program pulse	t_4	10	15	20	µs
Duration of output program pulse	t_5	15	20	20	µs
Time from end of program pulse to chip select (enable) low	t_6	5	5	10	µs
Time from chip select (enable) low to V_{CC} = 0 V	t_7	0.1	5	5	µs
Time for cooling between bits	t_8	30	50	100	µs
Time for cooling between words	t_9	30	50		µs

1/ T_C = +25°C.

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TABLE IVI. Programming characteristics for circuit I.

Parameter	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
V_{CC} during programming	V_{CCP}		5.0		5.5	V
High level input voltage during programming	V_{IHP}		2.4		5.5	V
Low level input voltage during programming	V_{ILP}		0.0		0.45	V
Chip enable voltage during programming	V_{CEP}	$\overline{CE}1$ pin	14.5		15.5	V
Output voltage during programming	V_{OP}		19.5		20.5	V
Voltage on outputs not to be programmed	V_{ONP}		0		V_{CCP} +0.3	V
Current on outputs not to be programmed	I_{ONP}				20	mA
Rate of output voltage change	$\frac{d(V_{OP})}{dt}$		20		250	V/µs
Rate of chip enable voltage change	$\frac{d(V_{CE})}{dt}$	$\overline{CE}1$ pin	100		1000	V/µs
Programming period	tp		50		100	µs
V_{CC} during programming verification	V_{CCL}		4.5		5.0	µs

1/ T_C = +25°C.

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TABLE IVJ. Programming characteristics for circuit J. 1/

Parameter	Symbol	Limits 1/			Unit
		Min	Recommended	Max	
Address input voltage 2/	V_{IH} V_{IL}	2.4 0.0	5.0 0.4	5.0 0.8	V V
Programming/verify voltage to V_{CC}	V_{PH} V_{PL}	11.75 4.5	12.0 4.5	12.25 5.5	V V
Programming voltage current limit with V_{PH} applied	I_{CCP}	600	600	650	mA
Voltage rise and fall time	t_r t_f	1.0 1.0	1.0 1.0	10 10	µs µs
Programming delay	t_d	10	10	100	µs
Programming pulse width	t_p	100		1000	µs
Programming duty cycle	DC	---	50	90	%
Output voltage Enable Disable 3/	 V_{OPE} V_{OPD}	 10.0 4.5	 10.5 5.0	 11.0 5.5	 V V

1/ T_C = +25°C.

2/ Address and chip select shall not be left open for V_{IH}.

3/ Disable condition shall be met with output open circuit.

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6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for programming the device, including processing option.

i. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Ground zero voltage potential.

V_{IN} - - - - - - - - - - - - - - - Voltage level at an input terminal.

V_{IC} - - - - - - - - - - - - - - - Input clamp voltage.

I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number. It is intended that spare devices for logistic support be acquired in the unprogrammed condition (see 3.7.1) and programmed by the maintenance activity, except where use quantities for devices with a specific program or pattern justify stocking of preprogrammed devices.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type/manufacturer	Circuit designator	Fusible link	Symbol/ FSCM number
01 6/	76160 /Harris	A	NiCr	CDWD/34371
01 6/	53S1680/Monolithic Memories	B	TiW	CECD/50364
01 6/	82S190 /Signetics Corp.	C	NiCr	CDKB/18324
01	77S190 /National	G	TiW/W	CCXP/27014
02 6/	76161 /Harris	A	NiCr	---
02	53S1681/Monolithic Memories	B	TiW	---
02,04	82S191A/Signetics Corp.	C	NiCr	---
02	3636 /Intel	E	Polysilicon	CECC/34649
02	29681 /Raytheon	F	NiCr	CRP/07933
02	77S191 /National	G	TiW/W	---
02,04	28S166A/Texas Instrument	H	TiW	CGO/01295
02	27S191 /Advanced Micro Devices	I	Platinum silicide	CDWN/34335
02	76161 /Motorola	J	NiCr	CGG/04713
03	93Z510 /Fairchild	D	ZVE 7/	CFJ/07263
04,02	93Z511 /Fairchild	D	ZVE	---
05, 6/	76165 /Harris	A	NiCr	---
05	82HS195/Signetics Corp.	C	ZVE	---

6/ This generic-industry types is no longer manufactured.

7/ Zapped vertical emitter.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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